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Abstract—This document describes the most common article elements and how to use the IEEEtran class with L^AT_EX to produce files that are suitable for submission to the IEEE. IEEEtran can produce conference, journal, and technical note (correspondence) papers with a suitable choice of class options.

Index Terms—Article submission, IEEE, IEEEtran, journal, L^AT_EX, paper, template, typesetting.

I. INTRODUCTION

Reconfigurable intelligent surfaces (RISs) have been recognized as a promising technology for future wireless networks because they can reshape the wireless propagation environment by imposing programmable electromagnetic responses on incident waves [1], [2]. By deploying a large number of low-cost passive or nearly passive reflecting elements, RISs can enhance coverage, improve link reliability, and create additional controllable propagation paths. However, these benefits critically rely on accurate channel state information (CSI). Since conventional passive RIS elements have limited or no radio-frequency chains, the RIS-related channels cannot be directly observed, and the acquisition of cascaded CSI becomes a major bottleneck in RIS-assisted systems [3]–[5].

To reduce the training overhead, many existing works exploit the cascaded structure of RIS channels. Early studies modeled the RIS-assisted channel as a bilinear or matrix-calibration problem and developed pilot designs, sparse recovery algorithms, or matrix factorization methods for estimating the user–RIS and RIS–BS components [3]–[5]. For wideband and multi-antenna systems, tensor decomposition provides a more structured alternative. In particular, the received pilot signals can be arranged as multiway arrays, and canonical polyadic decomposition (CPD) or PARAFAC-based methods can be used to exploit the multilinear structure induced by antennas, subcarriers, and RIS training blocks [6]. These tensor-based methods are attractive because they offer algebraic identifiability and can reduce the training overhead compared with unstructured least-squares estimation.

Despite these advances, most existing RIS tensor channel estimation models are not sufficient for the mixed-field distributed RIS-EVS-OFDM scenario considered in this paper. First, many tensor-based RIS estimators rely on far-field or plane-wave steering models. Such models are no longer accurate when the RIS aperture is electrically large and the user is located in the near-field region. In this case, the UE–RIS response contains spherical-wave phase curvature, and

the distance becomes an estimable channel parameter rather than a nuisance scaling term. Recent near-field XL-MIMO and XL-RIS studies have shown that spherical-wave propagation introduces new angle–distance coupling and requires dedicated near-field channel estimation or beam training designs [7], [8]. However, these methods are mostly developed in polar-domain or sparse-dictionary frameworks and do not directly address the heterogeneous tensor structure induced by distributed RIS training, EVS polarization, and OFDM frequency smoothing.

Second, the RIS training-domain factor is often treated as if it directly obeyed an array steering structure. This is not generally correct. Let $\mathbf{g}_k$ denote the RIS element-domain cascaded response of the k -th path and let $\mathbf{\Omega}_k$ denote the RIS training matrix. The observable fourth-mode factor in a tensor model is

$$\mathbf{c}_k = \mathbf{\Omega}_k \mathbf{g}_k, \quad (1)$$

not $\mathbf{g}_k$ itself. Therefore, the near-field spherical or Fresnel structure is imposed on the element-domain response $\mathbf{g}_k$, while the tensor factor is the compressed training-domain vector $\mathbf{c}_k$. Directly imposing an element-domain Hankel, Vandermonde, or near-field structure on $\mathbf{c}_k$ ignores the training compression and can lead to an inconsistent physical model. This distinction becomes particularly important when $T < M_R$, where T is the number of RIS training blocks and M_R is the number of RIS elements.

Third, existing tensor RIS estimators usually treat the spatial, delay, and RIS-training factors as algebraic components without fully enforcing their physical coupling. In a distributed RIS system illuminated by a single UE, the local UE–RIS ranges and angles associated with different RIS panels are not independent parameters. They are all induced by the same UE position and by a common clock offset. Similarly, the macroscopic OFDM delay of the k -th cascaded path satisfies

$$\tau_k = \frac{r_k + d_{\text{RB},k}}{c} + \Delta t, \quad (2)$$

where r_k is the UE–RIS distance, $d_{\text{RB},k}$ is the known RIS–BS distance, and Δt is the common clock offset. Treating τ_k , r_k , and the local angles as mutually independent variables causes a mismatch between the algebraic tensor model and the underlying single-UE geometry.

Fourth, conventional antenna-array channel models do not exploit the full electromagnetic information available at an electromagnetic vector sensor (EVS) array. An EVS measures multiple electric- and magnetic-field components, and thus provides polarization information in addition to spatial phase progression. Classical vector-sensor array processing has shown that EVSs can jointly characterize direction and polarization by using Maxwell-consistent vector responses

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[9]. Nevertheless, RIS tensor channel estimation has rarely incorporated EVS spatial-polarization structure into the tensor factorization process.

Motivated by these observations, this paper develops a mixed near/far-field distributed RIS-EVS-OFDM channel model and a heterogeneous physical-domain rank-one projected tensor estimation framework. The proposed model explicitly separates the UE-RIS near-field spherical response, the RIS-BS far-field response, the EVS spatial-polarization response, the OFDM delay response, and the RIS training compression. In particular, the RIS element-domain response is written as

$$\mathbf{g}_k = \mathbf{a}_{\text{RB},k} \odot \mathbf{a}_{\text{UR},k}^{\text{NF}}, \quad (3)$$

whereas the observable RIS training-domain tensor factor is

$$\mathbf{c}_k = \mathbf{\Omega}_k \mathbf{g}_k. \quad (4)$$

This distinction is essential: the near-field spherical or Fresnel structure is imposed on $\mathbf{g}_k$ in the RIS element domain, while the CPD factor observed from the tensor model is the compressed training-domain vector $\mathbf{c}_k$. Therefore, directly imposing a Vandermonde, Hankel, or near-field array structure on $\mathbf{c}_k$ would ignore the RIS training compression and may lead to a physically inconsistent estimator.

By applying frequency-domain Hankelization to the raw OFDM observations, the received pilot signal is arranged as a fourth-order tensor

$$\mathcal{Z} = \sum_{k=1}^K \beta_k \mathbf{a}_{\text{EVS},k} \circ \mathbf{b}_k \circ \mathbf{q}_k \circ \mathbf{c}_k + \mathcal{W}_H, \quad (5)$$

where $\mathbf{a}_{\text{EVS},k}$ is the BS-side EVS spatial-polarization factor, $\mathbf{b}_k$ and $\mathbf{q}_k$ are two frequency-smoothing Vandermonde factors sharing the same delay pole, and $\mathbf{c}_k$ is the compressed RIS near-field factor. Unlike conventional tensor models that treat all CPD factors as generic low-rank components, the proposed formulation associates different tensor modes with different physical generation mechanisms: an EVS-Kronecker-Maxwell manifold, a common-pole dual-delay Vandermonde manifold, and a compressed RIS near-field manifold.

The proposed estimator is inspired by the rank-one enhancement principle of exponential-structured CPD, where a Vandermonde factor can be mapped into a Hankel-domain rank-one representation. However, the considered RIS-EVS-OFDM tensor is fundamentally heterogeneous: not all factors are Vandermonde factors, and the observable RIS factor is $\mathbf{c}_k = \mathbf{\Omega}_k \mathbf{g}_k$ rather than the element-domain near-field response $\mathbf{g}_k$. Therefore, a direct ES-CPD-style Hankel projection is not valid for the RIS mode. To address this issue, we develop a heterogeneous physical-domain rank-one projection principle, where each mode is lifted according to its own physical structure. Specifically, the EVS factor is projected through a Maxwell-consistent Kronecker structure, the two delay factors are jointly projected through a common-pole ES-style Hankel rank-one lifting, and the RIS factor is projected through a compressed near-field dechirped rank-one lifting.

Based on this principle, we propose a structured estimation algorithm termed heterogeneous physical-domain rank-one projected CPD (HP-R1P-CPD). The algorithm consists of

three stages. First, a four-dimensional HOSVD-ESPRIT initialization extracts stable delay poles and initial tensor factors from the Hankelized tensor. Second, HP-R1P-CPD alternates between regularized least-squares factor updates and mode-specific physical projections. The mode-1 update projects the EVS factor onto the Maxwell-consistent spatial-polarization manifold. The mode-2 and mode-3 updates jointly enforce the common-delay pole shared by the two Hankelized OFDM factors through a mother delay factor. The mode-4 update solves a compressed near-field dechirped rank-one projection problem, where the Fresnel-dechirped rank-one structure is imposed in the RIS element domain and embedded into the compressed training-domain fitting of $\mathbf{c}_k$. An exact-spherical local refinement is further used as a physical safeguard to reduce the approximation bias of the Fresnel model. Third, the path-local tensor estimates are refined by a raw-domain global VP-WNLS estimator, where the nonlinear parameters are the UE position, the common clock offset, and the polarization parameters, while the complex path gains are eliminated by variable projection.

The main contributions of this paper are summarized as follows.

- We establish a mixed near/far-field distributed RIS-EVS-OFDM channel model. The proposed model explicitly distinguishes the RIS element-domain near-field response $\mathbf{g}_k$ from the compressed training-domain tensor factor $\mathbf{c}_k = \mathbf{\Omega}_k \mathbf{g}_k$, thereby avoiding the physically inconsistent practice of imposing element-domain near-field structure directly on the compressed RIS factor.
- We construct a fourth-order heterogeneous tensor observation model by combining EVS spatial-polarization sensing, OFDM frequency Hankelization, and RIS training-domain compression. The resulting tensor factors respectively follow an EVS-Kronecker-Maxwell structure, a common-pole dual-delay Vandermonde structure, and a compressed RIS near-field structure.
- We develop a physical-domain rank-one lifting framework that generalizes the Hankel-domain rank-one principle of exponential-structured CPD to heterogeneous wireless tensor factors. Different from directly applying ES-CPD to all modes, the proposed framework uses mode-specific liftings: Maxwell Kronecker lifting for the EVS factor, common-pole ES-style Hankel lifting for the dual delay factors, and Fresnel-dechirped block-Hankel lifting for the RIS near-field response.
- We propose an HP-R1P-CPD algorithm for structured tensor refinement. The algorithm alternates between regularized least-squares updates and physical-domain rank-one projections, thereby exploiting the electromagnetic, delay, and near-field structures of different tensor modes. The delay projection jointly enforces the common pole shared by $\mathbf{b}_k$ and $\mathbf{q}_k$, while the RIS projection maintains consistency with the compressed observation $\mathbf{c}_k = \mathbf{\Omega}_k \mathbf{g}_k$.
- We develop a compressed near-field dechirped rank-one projection for the RIS mode. Since the near-field structure exists in $\mathbf{g}_k$ rather than directly in $\mathbf{c}_k$, the proposed projection constructs a curvature-dependent dechirped

lifting in the RIS element domain and embeds it into a compressed-domain fitting problem. An exact-spherical local refinement is further incorporated to improve robustness when the Fresnel approximation is inaccurate.

- We refine the path-local tensor estimates by a raw-domain global VP-WNLS estimator. This final stage enforces that all local RIS-side ranges, angles, and macroscopic delays are induced by a single UE position and a common clock offset. Since Hankelization duplicates raw subcarrier samples and induces correlated tensor-domain noise, the final likelihood-based refinement is formulated in the original OFDM observation domain.

The rest of this paper is organized as follows. Section II presents the mixed near/far-field RIS-EVS-OFDM channel model and the fourth-order tensor observation model. Section III develops the heterogeneous physical-domain rank-one lifting structures, including the common-pole delay lifting, the EVS Maxwell-Kronecker lifting, and the compressed near-field dechirped lifting. Section IV proposes the HP-RIP-CPD algorithm and the raw-domain global VP-WNLS refinement. Section V discusses identifiability, the raw-domain Cramér-Rao bound, and computational complexity. Section VI provides numerical results, including baseline comparisons, ablation studies, robustness tests, training-overhead evaluation, and runtime scalability. Section VII concludes this paper.

II. SYSTEM AND CHANNEL MODEL

A. System Topology and Mixed-Field Propagation

We consider an uplink wideband OFDM training system assisted by K distributed RIS panels. The user equipment (UE) is equipped with a single antenna, while the base station (BS) employs a uniform linear array (ULA) of M_A electromagnetic vector sensors (EVSs). Since each EVS measures three electric-field and three magnetic-field components, the BS-side observation dimension is

$$I = 6M_A. \quad (6)$$

The center of the k -th RIS panel is denoted by $\mathbf{s}_k \in \mathbb{R}^3$, and the BS reference point is denoted by $\mathbf{p}_B \in \mathbb{R}^3$. The positions and orientations of all RIS panels and the BS are assumed to be known infrastructure information.

We focus on a dominant cascaded component associated with each distributed RIS panel. Specifically, the k -th dominant path propagates through the UE-RIS _{k} -BS cascade. The UE-RIS _{k} hop is modeled as a near-field spherical-wave link, while the RIS _{k} -BS hop is modeled as a far-field link due to the fixed and relatively large RIS-BS separation. Weak diffuse scattering and residual unmodeled components are treated as model mismatch or additive disturbance.

B. Local and Global Parameterizations

Let $\mathbf{p}_u \in \mathbb{R}^3$ denote the unknown UE position, and let Δt denote the common clock offset. For the k -th RIS, let $\mathbf{R}_k^{G \rightarrow R}$ be the known rotation matrix from the global coordinate system to the local coordinate system of RIS _{k} . The UE position in the local coordinate system of RIS _{k} is

$$\mathbf{q}_k = \mathbf{R}_k^{G \rightarrow R}(\mathbf{p}_u - \mathbf{s}_k), \quad (7)$$

and the corresponding UE-RIS distance and incident unit direction are

$$r_k = \|\mathbf{q}_k\|_2, \quad \mathbf{u}_k = \frac{\mathbf{q}_k}{r_k}. \quad (8)$$

The local elevation and azimuth angles at RIS _{k} are denoted by $(\vartheta_k^R, \varphi_k^R)$ and are determined by $\mathbf{u}_k$.

The RIS-BS center-to-center distance is known and given by

$$d_{RB,k} = \|\mathbf{s}_k - \mathbf{p}_B\|_2. \quad (9)$$

Therefore, the macroscopic delay of the k -th cascaded path is

$$\tau_k = \frac{r_k + d_{RB,k}}{c} + \Delta t, \quad (10)$$

where c is the speed of light. Hence, τ_k is not an independent physical parameter once $\mathbf{p}_u$ and Δt are specified.

For tensor-based estimation, we first use a path-local relaxed parameterization

$$\boldsymbol{\xi}_k = \{r_k, \vartheta_k^R, \varphi_k^R, \gamma_k, \eta_k\}, \quad (11)$$

where γ_k and η_k are the polarization auxiliary angle and polarization phase difference. The final global physical parameter vector is

$$\boldsymbol{\chi} = [\mathbf{p}_u^T, \Delta t, \gamma_1, \dots, \gamma_K, \eta_1, \dots, \eta_K]^T. \quad (12)$$

The local quantities r_k , ϑ_k^R , φ_k^R , and τ_k are derived from $\boldsymbol{\chi}$ in the final global refinement stage.

C. RIS Element-Domain Near-Field Response

Each RIS panel contains M_R reflecting elements. Let $\boldsymbol{\rho}_{k,m} \in \mathbb{R}^3$ denote the local coordinate of the m -th element of RIS _{k} with respect to the RIS center. Under the phase-dominant spherical-wave model, the distance difference between the m -th element and the RIS center is

$$\delta_{k,m} = \|\mathbf{q}_k - \boldsymbol{\rho}_{k,m}\|_2 - r_k. \quad (13)$$

The UE-RIS near-field steering vector is then defined as

$$[\mathbf{a}_{UR,k}^{\text{NF}}]_m = \exp\{-j\kappa\delta_{k,m}\}, \quad \kappa = \frac{2\pi}{\lambda_c}, \quad (14)$$

where $\lambda_c = c/f_c$ is the carrier wavelength. The common propagation phase associated with r_k is included in the macroscopic OFDM delay in (10), while (14) only retains the element-dependent spherical phase curvature.

The RIS-BS element-domain far-field response of RIS _{k} is denoted by $\mathbf{a}_{RB,k} \in \mathbb{C}^{M_R}$. Since the RIS and BS positions are known, $\mathbf{a}_{RB,k}$ is treated as a known deterministic response. The cascaded RIS element-domain response is

$$\mathbf{g}_k = \mathbf{a}_{RB,k} \odot \mathbf{a}_{UR,k}^{\text{NF}} \in \mathbb{C}^{M_R}, \quad (15)$$

where $\odot$ denotes the Hadamard product.

During the training phase, RIS _{k} applies T known phase-control patterns collected in

$$\boldsymbol{\Omega}_k = [\boldsymbol{\omega}_{k,1}, \dots, \boldsymbol{\omega}_{k,T}]^T \in \mathbb{C}^{T \times M_R}. \quad (16)$$

The corresponding compressed RIS training-domain factor is

$$\mathbf{c}_k = \boldsymbol{\Omega}_k \mathbf{g}_k \in \mathbb{C}^T. \quad (17)$$

It is important to distinguish the element-domain response $\mathbf{g}_k$ from the training-domain factor $\mathbf{c}_k$. The near-field spherical structure is imposed on $\mathbf{g}_k$, whereas the tensor factor observed in the fourth mode is the compressed vector $\mathbf{c}_k$.

The element-wise amplitude variation within one RIS panel is neglected and absorbed into the complex path gain. This phase-dominant approximation is accurate when the panel aperture is moderate compared with the UE-RIS distance. Including element-wise amplitude factors does not change the tensor structure, but is omitted for analytical clarity.

D. BS-Side EVS Spatial-Polarization Response

The RIS_k-BS far-field direction is known from the infras-structure geometry and is denoted by (θ_k^B, ϕ_k^B) . Let $\mathbf{v}_{B,k} \in \mathbb{C}^{M_A}$ be the BS ULA steering vector corresponding to this direction. The six-dimensional EVS polarization response is modeled through the direction-dependent Maxwell matrix

$$\mathbf{\Theta}_k = \mathbf{\Theta}(\theta_k^B, \phi_k^B) \in \mathbb{C}^{6 \times 2}. \quad (18)$$

We use the following two-dimensional polarization vector:

$$\mathbf{e}(\gamma_k, \eta_k) = \begin{bmatrix} \sin \gamma_k e^{j\eta_k} \\ \cos \gamma_k \end{bmatrix}. \quad (19)$$

The EVS polarization vector and the BS spatial-polarization steering vector are therefore

$$\mathbf{p}_k = \mathbf{\Theta}_k \mathbf{e}(\gamma_k, \eta_k) \in \mathbb{C}^6, \quad (20)$$

and

$$\mathbf{a}_{\text{EVS},k} = \mathbf{v}_{B,k} \otimes \mathbf{p}_k \in \mathbb{C}^{6M_A}, \quad (21)$$

where $\otimes$ denotes the Kronecker product.

E. Wideband OFDM Observation Model

The OFDM system contains N subcarriers with subcarrier spacing Δf . For the k -th cascaded path, the frequency-domain delay response is

$$d_k[n] = \exp\{-j2\pi n \Delta f \tau_k\}, \quad n = 0, \dots, N-1. \quad (22)$$

Let $\beta_k \in \mathbb{C}$ denote the complex path gain, which absorbs path loss, RIS reflection loss, residual common phase, and other scalar fading effects.

At subcarrier n and training block t , the received BS vector is

$$\mathbf{y}[n, t] = \sum_{k=1}^K \beta_k \mathbf{a}_{\text{EVS},k} d_k[n] \mathbf{c}_k[t] + \mathbf{w}[n, t], \quad (23)$$

where $\mathbf{w}[n, t] \sim \mathcal{CN}(\mathbf{0}, \sigma^2 \mathbf{I}_I)$ is the additive noise in the original OFDM observation domain. Collecting all subcarriers and training blocks yields the raw observation tensor

$$\mathcal{Y} \in \mathbb{C}^{I \times N \times T}. \quad (24)$$

F. Spatial Narrowband Assumption

Although the system is wideband in the OFDM delay domain, the spatial and polarization responses are assumed to be frequency-flat across the signal bandwidth. This spatial narrowband assumption requires that the fractional bandwidth is small and that the maximum array-induced differential delay over the BS array and each RIS panel is negligible compared with the inverse signal bandwidth. Under this condition, the wideband effect is captured by the scalar OFDM delay response in (22), while $\mathbf{a}_{\text{EVS},k}$ and $\mathbf{g}_k$ can be evaluated at the carrier frequency.

G. Frequency Hankelization and Four-Dimensional Tensor Model

To expose the Vandermonde structure of the delay response, frequency-domain Hankelization is applied to the raw OFDM observation. Let P be the smoothing window length and $L = N - P + 1$ be the number of sliding windows. The fourth-order Hankelized tensor

$$\mathcal{Z} \in \mathbb{C}^{I \times P \times L \times T} \quad (25)$$

is defined as

$$[\mathcal{Z}]_{i,p,\ell,t} = [\mathcal{Y}]_{i,p+\ell-1,t}, \quad p = 1, \dots, P, \quad \ell = 1, \dots, L. \quad (26)$$

For the k -th path, define the common delay pole

$$z_k = \exp\{-j2\pi \Delta f \tau_k\}. \quad (27)$$

Then the two frequency-smoothed delay factors are

$$\mathbf{b}_k = [1, z_k, \dots, z_k^{P-1}]^T \in \mathbb{C}^P, \quad (28)$$

and

$$\mathbf{q}_k = [1, z_k, \dots, z_k^{L-1}]^T \in \mathbb{C}^L. \quad (29)$$

Substituting (23) into (26) gives the following fourth-order CP model:

$$\mathcal{Z} = \sum_{k=1}^K \beta_k \mathbf{a}_{\text{EVS},k} \circ \mathbf{b}_k \circ \mathbf{q}_k \circ \mathbf{c}_k + \mathcal{W}_H, \quad (30)$$

where $\circ$ denotes the vector outer product and $\mathcal{W}_H$ is the Hankelized noise tensor.

Equivalently,

$$\mathcal{Z} = [[\beta; \mathbf{A}, \mathbf{B}, \mathbf{Q}, \mathbf{C}]] + \mathcal{W}_H, \quad (31)$$

where

$$\mathbf{A} = [\mathbf{a}_{\text{EVS},1}, \dots, \mathbf{a}_{\text{EVS},K}], \quad (32)$$

$$\mathbf{B} = [\mathbf{b}_1, \dots, \mathbf{b}_K], \quad (33)$$

$$\mathbf{Q} = [\mathbf{q}_1, \dots, \mathbf{q}_K], \quad (34)$$

$$\mathbf{C} = [\mathbf{c}_1, \dots, \mathbf{c}_K], \quad (35)$$

$$\boldsymbol{\beta} = [\beta_1, \dots, \beta_K]^T. \quad (36)$$

The Hankelized tensor $\mathcal{Z}$ is used to exploit the multi-way low-rank structure for initialization and structured factor refinement. However, Hankelization repeats the same raw subcarrier samples across different anti-diagonals. Therefore, the entries of $\mathcal{W}_H$ are correlated even when the raw-domain

noise in (23) is white. The final likelihood-based refinement and the CRB analysis are consequently formulated in the raw OFDM domain $\mathcal{Y}$ rather than by treating the entries of $\mathcal{Z}$ as independent observations.

H. Summary of Structured Manifolds

The model in (30) contains four heterogeneous physical structures:

- **Mode-1:** $\mathbf{a}_{\text{EVS},k}$ follows the BS spatial-polarization Kronecker structure with Maxwell-consistent EVS polarization response.
- **Mode-2 and Mode-3:** $\mathbf{b}_k$ and $\mathbf{q}_k$ share the same delay pole z_k , and hence the same macroscopic delay τ_k .
- **Mode-4:** $\mathbf{c}_k$ is the compressed RIS training-domain factor generated by the element-domain near-field response $\mathbf{g}_k$ through $\mathbf{c}_k = \mathbf{\Omega}_k \mathbf{g}_k$.
- **Global geometry:** the local quantities r_k , ϑ_k^R , φ_k^R , and τ_k are all induced by the common UE position $\mathbf{p}_u$ and the common clock offset Δt .

These coupled structures provide the physical basis for the proposed geometry-coupled structured tensor estimation algorithm.

III. HETEROGENEOUS PHYSICAL-DOMAIN RANK-ONE LIFTINGS FOR STRUCTURED TENSOR PROJECTION

This section develops the physical-domain rank-one lifting structures that support the proposed structured tensor refinement. The key idea is to convert the heterogeneous physical constraints embedded in the tensor factors into rank-one constraints in properly lifted domains. Different from a conventional CPD refinement, where the factor matrices are treated as generic low-rank algebraic components, the proposed formulation exploits the fact that the four tensor modes in (??) are generated by different propagation mechanisms. The EVS factor follows a Maxwell-consistent spatial-polarization structure, the two frequency-smoothed delay factors share the same delay pole, and the RIS factor is a compressed training-domain observation of an element-domain near-field response. Therefore, a universal projection rule is not appropriate; instead, each tensor mode requires a lifting operator dictated by its own physical generation mechanism.

A. Motivation: From ES-CPD to Heterogeneous Physical Liftings

Exponential-structured CPD methods are based on the observation that a Vandermonde or single-exponential factor can be mapped into a Hankel-domain rank-one representation. Specifically, if $\mathbf{x} = [1, z, \dots, z^{N-1}]^T$ is a single complex exponential, then the Hankel matrix $\mathcal{H}(\mathbf{x})$ satisfies

$$\mathcal{H}(\mathbf{x}) = \mathbf{u}(z) \mathbf{v}^T(z), \quad (37)$$

which is exactly rank-one. This rank-one property enables structured factor updates by alternating between least-squares fitting in the original tensor domain and rank-one projection in the transformed Hankel domain.

The RIS-EVS-OFDM tensor considered in this paper is more heterogeneous than the standard exponential-structured CPD setting. The delay factors are Vandermonde factors and therefore can be handled by an ES-style Hankel lifting. However, the EVS factor is not a Vandermonde factor; it is a Kronecker product between a BS-side spatial steering vector and a Maxwell-consistent polarization vector. More importantly, the RIS near-field factor is not a far-field Vandermonde factor, and the observable fourth-mode tensor factor is

$$\mathbf{c}_k = \mathbf{\Omega}_k \mathbf{g}_k, \quad \mathbf{g}_k = \mathbf{a}_{\text{RB},k} \odot \mathbf{a}_{\text{UR},k}^{\text{NF}}, \quad (38)$$

rather than the element-domain near-field response $\mathbf{g}_k$ itself. Therefore, directly imposing a Hankel or Vandermonde rank-one constraint on $\mathbf{c}_k$ is physically invalid in general. This motivates a heterogeneous physical-domain rank-one lifting framework, where the delay factors are projected through a common-pole ES-style Hankel lifting, the EVS factor is projected through a Maxwell-Kronecker rank-one lifting, and the RIS factor is projected through a compressed near-field dechirped rank-one lifting.

B. General Physical-Domain Rank-One Projection Principle

For the m -th tensor mode, let $\mathbf{x}_m(\boldsymbol{\xi}_m) \in \mathbb{C}^{d_m}$ denote a physical factor parameterized by $\boldsymbol{\xi}_m$. A physical-domain rank-one lifting is a linear or structured lifting operator

$$\mathcal{L}_m : \mathbb{C}^{d_m} \rightarrow \mathbb{C}^{p_m \times q_m} \quad (39)$$

such that

$$\mathcal{L}_m(\mathbf{x}_m(\boldsymbol{\xi}_m)) = \mathbf{u}_m(\boldsymbol{\xi}_m) \mathbf{v}_m^T(\boldsymbol{\xi}_m) + \mathbf{E}_m(\boldsymbol{\xi}_m), \quad (40)$$

where $\mathbf{u}_m(\boldsymbol{\xi}_m) \in \mathbb{C}^{p_m}$ and $\mathbf{v}_m(\boldsymbol{\xi}_m) \in \mathbb{C}^{q_m}$ are lifted-domain physical factors, and $\mathbf{E}_m(\boldsymbol{\xi}_m)$ is the lifting residual. If $\mathbf{E}_m(\boldsymbol{\xi}_m) = \mathbf{0}$ for all admissible $\boldsymbol{\xi}_m$, the lifting is exact. If $\mathbf{E}_m(\boldsymbol{\xi}_m) \neq \mathbf{0}$ but its norm is controlled by a physically interpretable approximation error, the lifting is approximate.

Given an unconstrained least-squares estimate $\tilde{\mathbf{x}}_m$, the physical-domain rank-one projection (PR1P) consists of three conceptual steps:

$$\tilde{\mathbf{x}}_m \xrightarrow{\mathcal{L}_m} \mathcal{L}_m(\tilde{\mathbf{x}}_m) \xrightarrow{\Pi_{r1}} \widehat{\mathbf{M}}_m \xrightarrow{\mathcal{L}_m^\dagger} \hat{\mathbf{x}}_m, \quad (41)$$

where $\Pi_{r1}(\cdot)$ denotes the best rank-one approximation obtained by truncated SVD, and $\mathcal{L}_m^\dagger$ denotes the corresponding inverse or pseudo-inverse de-lifting operation. For an exact PR1P, this procedure is model-consistent and introduces no approximation bias. For an approximate PR1P, the bias is controlled by the residual term in (40). The lifting operator is not universal; it is dictated by the physical generation mechanism of each factor.

C. Common-Pole ES-Style Delay Lifting

The two frequency-smoothed OFDM factors $\mathbf{b}_k$ and $\mathbf{q}_k$ originate from the same physical macroscopic delay τ_k . Define the delay pole

$$z_k = \exp(-j2\pi\Delta f\tau_k), \quad (42)$$

where Δf is the subcarrier spacing. The two delay factors are $\mathbf{b}_k = [1, z_k, \dots, z_k^{P-1}]^T \in \mathbb{C}^P$, $\mathbf{q}_k = [1, z_k, \dots, z_k^{L-1}]^T \in \mathbb{C}^L$ (43)

Although $\mathbf{b}_k$ and $\mathbf{q}_k$ have different lengths, they must share the same pole z_k . Therefore, applying ES-style projection separately to $\mathbf{B}$ and $\mathbf{Q}$ is not sufficient, because independent projections may destroy the common-delay constraint.

To enforce this coupling, we introduce a mother delay factor

$$\mathbf{D}_d = [\mathbf{d}_1, \dots, \mathbf{d}_K] \in \mathbb{C}^{R \times K}, \quad R = \max(P, L), \quad (44)$$

where

$$\mathbf{d}_k = [1, z_k, \dots, z_k^{R-1}]^T. \quad (45)$$

Let $\mathbf{J}_P \in \mathbb{C}^{P \times R}$ and $\mathbf{J}_L \in \mathbb{C}^{L \times R}$ be row-selection matrices that extract the first P and L entries, respectively. Then

$$\mathbf{B} = \mathbf{J}_P \mathbf{D}_d, \quad \mathbf{Q} = \mathbf{J}_L \mathbf{D}_d. \quad (46)$$

The common-pole delay constraint is thus imposed on $\mathbf{D}_d$ rather than separately on $\mathbf{B}$ and $\mathbf{Q}$.

For each column $\mathbf{d}_k$, define the Hankel lifting

$$\mathcal{H}(\mathbf{d}_k) \in \mathbb{C}^{P_d \times L_d}, \quad [\mathcal{H}(\mathbf{d}_k)]_{p,\ell} = d_k[p + \ell - 2], \quad (47)$$

where $P_d + L_d - 1 \leq R$. Since $d_k[n] = z_k^n$, we have

$$[\mathcal{H}(\mathbf{d}_k)]_{p,\ell} = z_k^{p-1} z_k^{\ell-1}, \quad (48)$$

and hence

$$\mathcal{H}(\mathbf{d}_k) = \mathbf{u}_d(z_k) \mathbf{v}_d^T(z_k), \quad (49)$$

where $\mathbf{u}_d(z_k) = [1, z_k, \dots, z_k^{P_d-1}]^T$ and $\mathbf{v}_d(z_k) = [1, z_k, \dots, z_k^{L_d-1}]^T$. Therefore, the delay lifting is exact rank-one.

For the matrix $\mathbf{D}_d$, the columnwise Hankel lifting forms a third-order tensor

$$\mathcal{F}(\mathbf{D}_d) = [\mathcal{H}(\mathbf{d}_1), \dots, \mathcal{H}(\mathbf{d}_K)], \quad (50)$$

whose frontal slices are rank-one matrices. Denote the corresponding frontal-rank-one set by

$$\mathcal{T}_d = \{\mathcal{X} : \text{rank}(\mathcal{X}(:, :, k)) = 1, k = 1, \dots, K\}. \quad (51)$$

Then the common-pole delay manifold can be written as

$$\mathcal{F}(\mathbf{D}_d) \in \mathcal{T}_d, \quad \mathbf{B} = \mathbf{J}_P \mathbf{D}_d, \quad \mathbf{Q} = \mathbf{J}_L \mathbf{D}_d. \quad (52)$$

Unlike applying ES-CPD separately to $\mathbf{B}$ and $\mathbf{Q}$, the proposed lifting is imposed on the mother delay factor $\mathbf{D}_d$, which enforces the common-pole constraint between the two Hankelized OFDM modes.

The common-pole delay lifting in this section defines the structured delay manifold rather than a complete update rule. In the proposed algorithm, $\mathbf{B}$ and $\mathbf{Q}$ are not updated by independent least-squares steps followed by a post-processing projection. Instead, a mother delay factor $\mathbf{D}_d$ is updated directly through a common-pole ES-style structured LS subproblem. Since $\mathbf{B}$ and $\mathbf{Q}$ appear bilinearly in the CPD model, the resulting update is implemented as a Gauss–Seidel structured LS step: at each outer iteration, the counterpart delay factor and the other tensor factors are fixed to construct a linearized

delay subproblem in $\mathbf{D}_d$, and the Hankel-domain frontal-rank-one constraint $\mathcal{F}(\mathbf{D}_d) \in \mathcal{T}_d$ is enforced by projected gradient iterations. The detailed update is given in Section IV.

After the structured LS update in Section IV produces a projected mother factor, the unit-circle delay pole can be corrected columnwise as

$$s_k = \sum_{r=1}^{R-1} \hat{d}_{r+1,k} \hat{d}_{r,k}^*, \quad \hat{z}_k = \frac{s_k}{|s_k| + \epsilon}, \quad (53)$$

where $\epsilon > 0$ is a small regularization constant. The delay estimate is

$$\hat{\tau}_k = -\frac{\angle \hat{z}_k}{2\pi \Delta f}. \quad (54)$$

The delay is identifiable within the unambiguous range $0 \leq \tau_k < 1/\Delta f$, or equivalently after coarse synchronization removes integer ambiguities. The projected mother factor and its two selected delay factors are reconstructed as

$$\hat{\mathbf{d}}_k = [1, \hat{z}_k, \dots, \hat{z}_k^{R-1}]^T, \quad \hat{\mathbf{B}} = \mathbf{J}_P \hat{\mathbf{D}}_d, \quad \hat{\mathbf{Q}} = \mathbf{J}_L \hat{\mathbf{D}}_d. \quad (55)$$

D. EVS Maxwell-Kronecker Rank-One Lifting

The BS-side EVS factor is generated by the Kronecker product of a known or geometry-determined BS-side spatial response and a Maxwell-consistent polarization response. For the k -th path, we write

$$\mathbf{a}_{\text{EVS},k} = \mathbf{v}_{\text{B},k} \otimes \mathbf{p}_k \in \mathbb{C}^{6M_A}, \quad (56)$$

where $\mathbf{v}_{\text{B},k} \in \mathbb{C}^{M_A}$ denotes the BS-side spatial steering vector associated with the RIS-BS direction, and $\mathbf{p}_k \in \mathbb{C}^6$ denotes the EVS polarization response. The Maxwell-consistent polarization response is modeled as

$$\mathbf{p}_k = \mathbf{\Theta}_k \mathbf{e}(\gamma_k, \eta_k), \quad (57)$$

where $\mathbf{\Theta}_k \in \mathbb{C}^{6 \times 2}$ is determined by the propagation direction and the EVS electromagnetic response, and

$$\mathbf{e}(\gamma_k, \eta_k) = \begin{bmatrix} \sin \gamma_k e^{j\eta_k} \\ \cos \gamma_k \end{bmatrix}, \quad \gamma_k \in [0, \pi/2], \quad \eta_k \in (-\pi, \pi]. \quad (58)$$

The above phase convention fixes the second component of $\mathbf{e}(\gamma_k, \eta_k)$ as real and nonnegative, while the remaining global phase is absorbed into the complex path gain.

Define the reshaping operator

$$\text{mat}_{6 \times M_A}(\mathbf{a}_{\text{EVS},k}) = \mathbf{A}_{\text{EVS},k}. \quad (59)$$

Using (56), we obtain

$$\mathbf{A}_{\text{EVS},k} = \mathbf{p}_k \mathbf{v}_{\text{B},k}^T = \mathbf{\Theta}_k \mathbf{e}(\gamma_k, \eta_k) \mathbf{v}_{\text{B},k}^T. \quad (60)$$

Therefore, the EVS lifting

$$\mathcal{L}_{\text{EVS}}(\mathbf{a}_{\text{EVS},k}) = \text{mat}_{6 \times M_A}(\mathbf{A}_{\text{EVS},k}) \quad (61)$$

is exact rank-one. This projection is not an ES-CPD-type Hankel lifting; it is a Maxwell-consistent Kronecker rank-one lifting.

Given an unconstrained estimate $\tilde{\mathbf{a}}_k$, let

$$\tilde{\mathbf{A}}_k = \text{mat}_{6 \times M_A}(\tilde{\mathbf{a}}_k). \quad (62)$$

If $\mathbf{v}_{B,k}$ is determined by known RIS-BS geometry, the polarization proxy can be obtained by removing the spatial steering component:

$$\tilde{\mathbf{p}}_k = \frac{\tilde{\mathbf{A}}_k \mathbf{v}_{B,k}^*}{\|\mathbf{v}_{B,k}\|_2^2 + \epsilon}. \quad (63)$$

The Maxwell-subspace projection is

$$\bar{\mathbf{p}}_k = \mathbf{P}_{\Theta_k} \tilde{\mathbf{p}}_k, \quad \mathbf{P}_{\Theta_k} = \Theta_k \left(\Theta_k^H \Theta_k + \epsilon \mathbf{I} \right)^{-1} \Theta_k^H. \quad (64)$$

Equivalently, the two-dimensional polarization coefficient is estimated as

$$\tilde{\mathbf{e}}_k = \left(\Theta_k^H \Theta_k + \epsilon \mathbf{I} \right)^{-1} \Theta_k^H \tilde{\mathbf{p}}_k. \quad (65)$$

To remove the global phase ambiguity, the dominant nonzero component is used as the phase reference. In the nominal case, the second component is used:

$$\bar{\mathbf{e}}_k = \tilde{\mathbf{e}}_k e^{-j\angle[\tilde{\mathbf{e}}_k]_2}. \quad (66)$$

If $|\angle[\tilde{\mathbf{e}}_k]_2|$ is close to zero, which may occur when $\gamma_k \approx \pi/2$, the above convention becomes numerically unstable; in that case, the first nonzero component is used as the phase reference instead. The projected EVS factor is then reconstructed as

$$\hat{\mathbf{a}}_{\text{EVS},k} = \mathbf{v}_{B,k} \otimes \Theta_k \hat{\mathbf{e}}_k, \quad (67)$$

where $\hat{\mathbf{e}}_k$ follows the phase convention in (58). The remaining scalar ambiguity is absorbed into the path gain.

E. RIS Grid and Two-Dimensional Block-Hankel Lifting

We next specify the RIS grid and the dimensions of the two-dimensional block-Hankel lifting used for the near-field mode. Consider a rectangular RIS panel with

$$M_R = M_x M_y \quad (68)$$

elements, where M_x and M_y are the numbers of elements along the horizontal and vertical axes. The position of the (m_x, m_y) -th RIS element relative to the panel center is defined as

$$\boldsymbol{\rho}_{m_x, m_y} = \begin{bmatrix} \left(m_x - \frac{M_x-1}{2} \right) d_x \\ \left(m_y - \frac{M_y-1}{2} \right) d_y \\ 0 \end{bmatrix}, \quad m_x = 0, \dots, M_x-1, \quad m_y = 0, \dots, M_y-1, \quad (69)$$

where d_x and d_y denote the inter-element spacings. For a vectorized RIS response $\mathbf{x} \in \mathbb{C}^{M_R}$, let $\mathbf{X}_R \in \mathbb{C}^{M_x \times M_y}$ denote its matrix form with $[\mathbf{X}_R]_{m_x+1, m_y+1} = x_{m_x, m_y}$ under the adopted vectorization order.

Choose the two-dimensional Hankel window sizes P_x, L_x, P_y, L_y such that

$$P_x + L_x - 1 = M_x, \quad P_y + L_y - 1 = M_y. \quad (70)$$

The two-dimensional block-Hankel lifting

$$\mathcal{H}_{2D} : \mathbb{C}^{M_x \times M_y} \rightarrow \mathbb{C}^{P_x P_y \times L_x L_y} \quad (71)$$

is defined by

$$[\mathcal{H}_{2D}(\mathbf{X}_R)]_{(p_x, p_y), (\ell_x, \ell_y)} = [\mathbf{X}_R]_{p_x + \ell_x - 1, p_y + \ell_y - 1}, \quad (72)$$

where $p_x = 1, \dots, P_x$, $\ell_x = 1, \dots, L_x$, $p_y = 1, \dots, P_y$, and $\ell_y = 1, \dots, L_y$. The row and column multi-indices (p_x, p_y) and (ℓ_x, ℓ_y) are mapped to scalar row and column indices in a fixed lexicographic order. The inverse operator $\mathcal{H}_{2D}^\dagger$ denotes two-dimensional anti-diagonal averaging:

$$[\mathcal{H}_{2D}^\dagger(\mathbf{X})]_{m_x, m_y} = \frac{1}{\rho_{m_x, m_y}} \sum_{\substack{p_x + \ell_x - 1 = m_x \\ p_y + \ell_y - 1 = m_y}} [\mathbf{X}]_{(p_x, p_y), (\ell_x, \ell_y)}, \quad (73)$$

where ρ_{m_x, m_y} is the number of entries contributing to the (m_x, m_y) -th anti-diagonal average. This definition fully specifies the dimensions of $\mathcal{H}_{2D}(\cdot)$, $\mathcal{H}_{2D}^\dagger(\cdot)$, the dechirping matrix, and the lifted rank-one variable used below.

F. Element-Domain Fresnel-Dechirped Near-Field Lifting

Let r_k be the UE-RIS distance and let $\mathbf{u}_k = \mathbf{u}(\vartheta_k, \varphi_k)$ be the unit direction vector from the RIS panel to the UE. Under the second-order Fresnel approximation, the UE-RIS path-length difference can be written as

$$\delta_{k,m}^F = -\mathbf{u}_k^T \boldsymbol{\rho}_m + \frac{\boldsymbol{\rho}_m^T (\mathbf{I} - \mathbf{u}_k \mathbf{u}_k^T) \boldsymbol{\rho}_m}{2r_k}. \quad (74)$$

Thus, the near-field RIS response is approximated by

$$[\mathbf{a}_{\text{UR},k}^{\text{NF},F}]_m = \exp(j\kappa \mathbf{u}_k^T \boldsymbol{\rho}_m - j\boldsymbol{\rho}_m^T \mathbf{C}_k \boldsymbol{\rho}_m), \quad (75)$$

where $\kappa = 2\pi/\lambda_c$ and

$$\mathbf{C}_k = \frac{\kappa}{2r_k} (\mathbf{I} - \mathbf{u}_k \mathbf{u}_k^T). \quad (76)$$

This model is valid for a regular rectangular RIS grid when the second-order Fresnel approximation is accurate over the RIS aperture.

Because of the quadratic phase term in (75), the direct block-Hankel lifting is not rank-one. The nonseparable cross term can be removed by a curvature-dependent dechirping kernel. Let $\boldsymbol{\rho}_{p_x, p_y}^{(r)}$ and $\boldsymbol{\rho}_{\ell_x, \ell_y}^{(c)}$ denote the row-side and column-side offsets induced by the block-Hankel construction. Define

$$[\mathbf{D}_{\mathbf{C}_k}]_{(p_x, p_y), (\ell_x, \ell_y)} = \exp(j2(\boldsymbol{\rho}_{p_x, p_y}^{(r)})^T \mathbf{C}_k \boldsymbol{\rho}_{\ell_x, \ell_y}^{(c)}). \quad (77)$$

Then the dechirped block-Hankel lifting satisfies

$$\mathcal{H}_{2D}^\dagger \mathcal{H}_{2D} \left(\mathbf{a}_{\text{UR},k}^{\text{NF},F} \right) = \mathbf{f}_r(\boldsymbol{\eta}_k) \mathbf{f}_c^T(\boldsymbol{\eta}_k), \quad (78)$$

where $\mathbf{a}_{\text{UR},k}^{\text{NF},F}$ is the $M_x \times M_y$ matrix form of $\mathbf{a}_{\text{UR},k}^{\text{NF},F}$, $\boldsymbol{\eta}_k = (r_k, \vartheta_k, \varphi_k)$, $\mathbf{f}_r(\boldsymbol{\eta}_k) \in \mathbb{C}^{P_x P_y}$, and $\mathbf{f}_c(\boldsymbol{\eta}_k) \in \mathbb{C}^{L_x L_y}$. Hence, the Fresnel near-field response admits an exact dechirped rank-one lifting in the element domain for fixed $\boldsymbol{\eta}_k$:

$$\mathbf{X}_{\boldsymbol{\eta}_k}^{\text{phys}} = \mathbf{f}_r(\boldsymbol{\eta}_k) \mathbf{f}_c^T(\boldsymbol{\eta}_k). \quad (79)$$

The exact spherical-wave response differs from the Fresnel response by higher-order residual terms. Let $\mathbf{a}_{\text{UR},k}^{\text{NF},\text{ex}}$ denote the exact spherical response. The normalized rank-one residual of the dechirped lifting is controlled by the Fresnel approximation error and typically increases with the RIS aperture and carrier frequency, while decreasing with the UE-RIS range. Therefore, the Fresnel-dechirped rank-one lifting should be interpreted as a low-dimensional structured surrogate for candidate generation and denoising, not as the final physical RIS response. The

final RIS-side response used by the algorithm is validated and locally refined by the exact spherical model in the compressed observation domain.

G. Compressed Near-Field Dechirped Rank-One Lifting

The observable fourth-mode tensor factor is not the RIS element-domain response but its compressed training-domain version:

$$\mathbf{c}_k = \mathbf{\Omega}_k \mathbf{g}_k, \quad \mathbf{g}_k = \mathbf{a}_{\text{RB},k} \odot \mathbf{a}_{\text{UR},k}^{\text{NF}}. \quad (80)$$

When $T < M_R$, the pseudo-inverse back-projection $\mathbf{\Omega}_k^\dagger \tilde{\mathbf{c}}_k$ is generally not a reliable estimate of $\mathbf{g}_k$. Therefore, the main projection should not be formulated as an element-domain recovery followed by a near-field projection. Instead, the structured fitting should be performed directly in the compressed training domain.

For a candidate local near-field geometry

$$\boldsymbol{\eta}_k = (r_k, \vartheta_k, \varphi_k) \in \mathcal{R}_k, \quad (81)$$

define $\mathbf{u}_k = \mathbf{u}(\vartheta_k, \varphi_k)$ and $\mathbf{C}_k(\boldsymbol{\eta}_k)$ according to (76). Let $\mathbf{X}_k \in \mathbb{C}^{P_x P_y \times L_x L_y}$ be the dechirped lifted-domain matrix satisfying $\text{rank}(\mathbf{X}_k) = 1$. Define the element-domain de-lifting operator

$$\mathcal{S}_{\boldsymbol{\eta}_k}(\mathbf{X}_k) = \text{vec} \left\{ \mathcal{H}_{2D}^\dagger (\mathbf{D}_{\mathbf{C}_k}^* \odot \mathbf{X}_k) \right\}, \quad (82)$$

where $\mathcal{H}_{2D}^\dagger(\cdot)$ denotes inverse block-Hankel averaging. In (82), the multiplication by $\mathbf{D}_{\mathbf{C}_k}^*$ restores the Fresnel chirp term, and the inverse block-Hankel averaging maps the lifted matrix back to the RIS element domain.

The compressed near-field dechirped lifting operator is then defined as

$$\mathcal{T}_{k,\boldsymbol{\eta}_k}(\mathbf{X}_k) = \mathbf{\Omega}_k \text{diag}(\mathbf{a}_{\text{RB},k}) \mathcal{S}_{\boldsymbol{\eta}_k}(\mathbf{X}_k). \quad (83)$$

A purely arbitrary rank-one variable $\mathbf{X}_k$ would over-relax the physical near-field manifold, because for a fixed $\boldsymbol{\eta}_k$ the Fresnel response corresponds to the specific rank-one matrix $\mathbf{X}_{\boldsymbol{\eta}_k}^{\text{phys}} = \mathbf{f}_r(\boldsymbol{\eta}_k) \mathbf{f}_c^T(\boldsymbol{\eta}_k)$ rather than an arbitrary rank-one matrix. To avoid this over-relaxation, the proposed projection uses $\mathbf{X}_k$ only as a structured surrogate for local denoising and candidate generation, and anchors it to the physically generated Fresnel rank-one matrix.

The physically regularized compressed near-field fitting for a fixed $\boldsymbol{\eta}_k$ is formulated as

$$\hat{\mathbf{X}}_k(\boldsymbol{\eta}_k) = \arg \min_{\text{rank}(\mathbf{X}_k)=1} \left\| \tilde{\mathbf{c}}_k - \mathcal{T}_{k,\boldsymbol{\eta}_k}(\mathbf{X}_k) \right\|_2^2 + \lambda_{\text{phys}} \left\| \mathbf{X}_k - \mathbf{X}_{\boldsymbol{\eta}_k}^{\text{phys}} \right\|_F^2, \quad (84)$$

where $\lambda_{\text{phys}} > 0$ controls how tightly the lifted variable is tied to the physical Fresnel response. In implementation, $\mathbf{X}_k$ is initialized by $\mathbf{X}_{\boldsymbol{\eta}_k}^{\text{phys}}$ and only a small number of projected-gradient steps is performed. Hence, the variable $\mathbf{X}_k$ serves as a local denoising surrogate rather than a free lifted-domain representation.

Let $\mathbf{x}_k = \text{vec}(\mathbf{X}_k)$, $\mathbf{x}_{\boldsymbol{\eta}_k}^{\text{phys}} = \text{vec}(\mathbf{X}_{\boldsymbol{\eta}_k}^{\text{phys}})$, and write the linear operator in matrix form as

$$\mathcal{T}_{k,\boldsymbol{\eta}_k}(\mathbf{X}_k) = \mathbf{T}_{k,\boldsymbol{\eta}_k} \mathbf{x}_k. \quad (85)$$

Then a projected-gradient step for fixed $\boldsymbol{\eta}_k$ is

$$\mathbf{x}_k^{(j+\frac{1}{2})} = \mathbf{x}_k^{(j)} + \mu_c \left[\mathbf{T}_{k,\boldsymbol{\eta}_k}^H \left(\tilde{\mathbf{c}}_k - \mathbf{T}_{k,\boldsymbol{\eta}_k} \mathbf{x}_k^{(j)} \right) - \lambda_{\text{phys}} \left(\mathbf{x}_k^{(j)} - \mathbf{x}_{\boldsymbol{\eta}_k}^{\text{phys}} \right) \right], \quad (86)$$

with

$$0 < \mu_c < \frac{1}{\|\mathbf{T}_{k,\boldsymbol{\eta}_k}\|_2^2 + \lambda_{\text{phys}}}. \quad (87)$$

The rank-one projection is then applied as

$$\mathbf{X}_k^{(j+\frac{1}{2})} = \text{mat} \left(\mathbf{x}_k^{(j+\frac{1}{2})} \right), \quad \mathbf{X}_k^{(j+1)} = \Pi_{\text{r1}} \left(\mathbf{X}_k^{(j+\frac{1}{2})} \right). \quad (88)$$

For each candidate $\boldsymbol{\eta}_k \in \mathcal{G}_k$, the regularized residual

$$\rho_k(\boldsymbol{\eta}_k) = \left\| \tilde{\mathbf{c}}_k - \mathcal{T}_{k,\boldsymbol{\eta}_k} \left(\hat{\mathbf{X}}_k(\boldsymbol{\eta}_k) \right) \right\|_2^2 + \lambda_{\text{phys}} \left\| \hat{\mathbf{X}}_k(\boldsymbol{\eta}_k) - \mathbf{X}_{\boldsymbol{\eta}_k}^{\text{phys}} \right\|_F^2 \quad (89)$$

is evaluated, and the Fresnel-domain candidate is selected as

$$\hat{\boldsymbol{\eta}}_k^{\text{F}} = \arg \min_{\boldsymbol{\eta}_k \in \mathcal{G}_k} \rho_k(\boldsymbol{\eta}_k). \quad (90)$$

The exact spherical model is used as a local physical refinement or compressed-domain validation to reduce the approximation bias of the Fresnel surrogate. Define

$$\mathbf{h}_k^{\text{ex}}(\boldsymbol{\eta}) = \mathbf{\Omega}_k \left[\mathbf{a}_{\text{RB},k} \odot \mathbf{a}_{\text{UR},k}^{\text{NF,ex}}(\boldsymbol{\eta}) \right]. \quad (91)$$

For a fixed $\boldsymbol{\eta}$, the optimal complex scale is

$$\alpha_k(\boldsymbol{\eta}) = \frac{(\mathbf{h}_k^{\text{ex}}(\boldsymbol{\eta}))^H \tilde{\mathbf{c}}_k}{\|\mathbf{h}_k^{\text{ex}}(\boldsymbol{\eta})\|_2^2 + \epsilon}. \quad (92)$$

The local exact-spherical objective is

$$\Phi_k(\boldsymbol{\eta}) = \|\tilde{\mathbf{c}}_k - \alpha_k(\boldsymbol{\eta}) \mathbf{h}_k^{\text{ex}}(\boldsymbol{\eta})\|_2^2. \quad (93)$$

Starting from $\hat{\boldsymbol{\eta}}_k^{\text{F}}$, a local refinement gives

$$\hat{\boldsymbol{\eta}}_k^{\text{ex}} = \text{LocalRefine} \left(\hat{\boldsymbol{\eta}}_k^{\text{F}}; \Phi_k(\boldsymbol{\eta}) \right). \quad (94)$$

The final fourth-mode factor is reconstructed as

$$\hat{\mathbf{c}}_k = \alpha_k(\hat{\boldsymbol{\eta}}_k^{\text{ex}}) \mathbf{h}_k^{\text{ex}}(\hat{\boldsymbol{\eta}}_k^{\text{ex}}), \quad \hat{\mathbf{c}}_k \leftarrow \frac{\hat{\mathbf{c}}_k}{\|\hat{\mathbf{c}}_k\|_2}. \quad (95)$$

This refinement should not be interpreted as a separate global exact-spherical projection. The Fresnel-dechirped rank-one lifting provides a low-dimensional structured candidate, while the exact-spherical step locally reduces the Fresnel approximation bias in the compressed observation domain.

H. Why Direct Hankel Projection on the Compressed RIS Factor Fails

The compressed RIS factor $\mathbf{c}_k$ generally does not inherit the Hankel rank-one property of the element-domain near-field response. To see this, suppose that the element-domain response $\mathbf{g}_k$ admits a rank-one representation after a certain dechirped block-Hankel lifting. The observed fourth-mode factor is

$$\mathbf{c}_k = \mathbf{\Omega}_k \mathbf{g}_k. \quad (96)$$

For a generic training matrix $\mathbf{\Omega}_k$, there is no identity of the form

$$\mathcal{H}(\mathbf{\Omega}_k \mathbf{g}_k) = \mathbf{A}_{\mathbf{\Omega}} \mathcal{H}(\mathbf{g}_k) \mathbf{B}_{\mathbf{\Omega}} \quad (97)$$

TABLE I
SUMMARY OF HETEROGENEOUS PHYSICAL-DOMAIN RANK-ONE LIFTINGS

Mode	Factor	Lifting	Rank-One Type	
Delay modes	$(\mathbf{B}, \mathbf{Q})$	Mother factor $\mathbf{D}_d$, with $\mathbf{B} = \mathbf{J}_P \mathbf{D}_d$ and $\mathbf{Q} = \mathbf{J}_L \mathbf{D}_d$	Common-pole Hankel rank-one	
EVS mode	$\mathbf{a}_{\text{EVS},k}$	$\text{mat}_{6 \times M_A}(\mathbf{a}_{\text{EVS},k})$	Maxwell-Kronecker rank-one	
RIS mode	$\mathbf{c}_k = \mathbf{\Omega}_k \mathbf{g}_k$	$\mathcal{T}_{k,\eta}(\mathbf{X})$ with $\text{rank}(\mathbf{X}) = 1$ and physical anchor $\mathbf{X}_\eta^{\text{phys}}$	Physically anchored compressed near-field dechirped rank-one fitting	Fr

that preserves rank-one structure. Consequently,

$$\text{rank}(\mathcal{H}(\mathbf{c}_k)) = 1 \quad (98)$$

does not hold in general, even if the corresponding dechirped element-domain lifting is rank-one. Such a property would require special commutation relations between the training compression and the Hankel lifting, which are not satisfied by general RIS training patterns.

This observation is important for model consistency. Directly applying an ES-CPD-style Hankel projection to $\mathbf{c}_k$ would impose a structure on the compressed training-domain vector that is not physically guaranteed. The proposed compressed near-field dechirped rank-one lifting in (83) resolves this issue by imposing the rank-one structure in the dechirped element-domain lifting while fitting the observable factor directly in the compressed training domain. The additional physical anchoring term in (84) further prevents the lifted variable from becoming an arbitrary rank-one matrix unrelated to the candidate Fresnel geometry.

I. Summary of Heterogeneous Liftings

The proposed PR1P framework maps different tensor factors to different rank-one domains. The delay factors use a common-pole ES-style Hankel lifting, the EVS factor uses a Maxwell-Kronecker lifting, and the RIS factor uses a physically anchored compressed near-field dechirped lifting. These liftings are summarized in Table I.

In summary, this section generalizes the transform-domain rank-one philosophy from Vandermonde factors to heterogeneous physical factors in RIS-EVS-OFDM channel estimation. The common-pole delay lifting preserves the shared delay pole between the two Hankelized OFDM modes. The EVS lifting enforces the Maxwell-consistent spatial-polarization Kronecker structure and includes a stable phase-reference convention for nearly degenerate polarization states. The compressed near-field lifting imposes the dechirped rank-one structure in the RIS element domain while fitting the observable fourth-mode factor directly in the RIS training domain, and the physical anchoring term prevents the lifted rank-one variable from becoming an over-relaxed nonphysical surrogate. These structures provide the theoretical basis for the HP-R1P-CPD algorithm developed in the next section.

IV. PROPOSED HP-R1P-CPD ALGORITHM AND RAW-DOMAIN GLOBAL REFINEMENT

This section develops the proposed heterogeneous physical-domain rank-one projected CPD (HP-R1P-CPD) algorithm for the mixed near/far-field RIS-EVS-OFDM tensor model. The algorithm uses the Hankelized tensor for initialization

and structured factor refinement, and then returns to the raw OFDM observation domain for final likelihood-based global refinement. This design is motivated by the fact that the Hankelized tensor exposes multilinear low-rank structure, whereas Hankelization duplicates raw subcarrier samples and induces correlated tensor-domain noise. Therefore, the structured CPD stage is used to obtain reliable path-local estimates, while the final VP-WNLS stage enforces the single-UE geometry in the original observation domain.

A. Algorithm Overview

Recall that the Hankelized tensor observation is modeled as

$$\mathcal{Z} = \sum_{k=1}^K \beta_k \mathbf{a}_k \circ \mathbf{b}_k \circ \mathbf{q}_k \circ \mathbf{c}_k + \mathcal{W}_H = [[\beta; \mathbf{A}, \mathbf{B}, \mathbf{Q}, \mathbf{C}]] + \mathcal{W}_H, \quad (99)$$

where $\mathbf{a}_k = \mathbf{a}_{\text{EVS},k}$, $\mathbf{b}_k = [1, z_k, \dots, z_k^{P-1}]^T$, $\mathbf{q}_k = [1, z_k, \dots, z_k^{L-1}]^T$, $z_k = e^{-j2\pi\Delta f\tau_k}$, and $\mathbf{c}_k = \mathbf{\Omega}_k \mathbf{g}_k$ with $\mathbf{g}_k = \mathbf{a}_{\text{RB},k} \odot \mathbf{a}_{\text{UR},k}^{\text{NF}}$. The key point is that $\mathbf{c}_k$ is a compressed training-domain factor, not the RIS element-domain near-field response itself. Hence, the RIS mode cannot be treated as an ordinary Vandermonde or directly Hankelized factor.

The proposed HP-R1P-CPD stage solves the following structured approximation problem:

$$\min_{\mathbf{A}, \mathbf{B}, \mathbf{Q}, \mathbf{C}, \beta} \|\mathcal{Z} - [[\beta; \mathbf{A}, \mathbf{B}, \mathbf{Q}, \mathbf{C}]]\|_F^2, \quad (100a)$$

$$\text{s.t. } \mathbf{A}(:, k) \in \mathcal{M}_{\text{EVS},k}, \quad (100b)$$

$$\mathbf{B}(:, k) = \mathbf{b}(z_k), \quad \mathbf{Q}(:, k) = \mathbf{q}(z_k), \quad z_k = e^{-j2\pi\Delta f\tau_k}, \quad (100c)$$

$$\mathbf{C}(:, k) \in \mathcal{M}_{\mathbf{\Omega}, \text{NF},k}, \quad k = 1, \dots, K. \quad (100d)$$

The constraint in (100b) corresponds to the EVS Maxwell-Kronecker manifold, (100c) enforces the common-pole dual-delay Vandermonde structure, and (100d) is the physically anchored compressed near-field manifold defined by the dechirped rank-one lifting in Section III. The output of (100) is still path-local, because the local ranges, angles, and delays are not yet forced to be induced by a single UE position. The final stage therefore estimates the global parameter vector

$$\boldsymbol{\chi} = [\mathbf{p}_u^T, \Delta t, \gamma_1, \dots, \gamma_K, \eta_1, \dots, \eta_K]^T \quad (101)$$

by raw-domain VP-WNLS, where the complex path gains are eliminated by variable projection.

B. Stage I: HOSVD-ESPRIT Initialization

The first stage provides a stable initialization for the structured alternating refinement. Directly solving (100) from random initial factors is sensitive to local minima because the delay, polarization, and compressed near-field structures are

nonlinear. We therefore initialize the factors by exploiting the multilinear low-rank structure of the Hankelized tensor.

We compute a truncated HOSVD of $\mathcal{Z}$:

$$\mathcal{Z} \approx \mathcal{S} \times_1 \mathbf{U}_A \times_2 \mathbf{U}_B \times_3 \mathbf{U}_Q \times_4 \mathbf{U}_C, \quad (102)$$

where the dominant mode ranks are chosen around the assumed number of paths K . Since modes 2 and 3 are generated by the same delay pole, their subspaces are used jointly to estimate the initial poles $\{z_k^{(0)}\}_{k=1}^K$ through TLS-ESPRIT or a joint shift-invariance operation. The initial delays are obtained as

$$\tau_k^{(0)} = -\frac{\angle z_k^{(0)}}{2\pi\Delta f}, \quad k = 1, \dots, K. \quad (103)$$

The initial delay factors are reconstructed by

$$\mathbf{b}_k^{(0)} = [1, z_k^{(0)}, \dots, (z_k^{(0)})^{P-1}]^T, \quad \mathbf{q}_k^{(0)} = [1, z_k^{(0)}, \dots, (z_k^{(0)})^{L-1}]^T. \quad (104)$$

With $\mathbf{B}^{(0)}$ and $\mathbf{Q}^{(0)}$ fixed, the coupled mode-(1,4) factor associated with the k -th path is estimated by regularized least squares from the corresponding unfolding of $\mathcal{Z}$. Denote this coupled factor by $\mathbf{h}_k^{(0)} \approx \mathbf{c}_k^{(0)} \otimes \mathbf{a}_k^{(0)}$. After reshaping $\mathbf{h}_k^{(0)}$ into an $I \times T$ matrix, a rank-one SVD yields

$$\text{mat}_{I \times T}(\mathbf{h}_k^{(0)}) \approx \mathbf{a}_k^{(0)} (\mathbf{c}_k^{(0)})^T. \quad (105)$$

The initial EVS factor $\mathbf{a}_k^{(0)}$ is projected onto the Maxwell-Kronecker manifold using the EVS projection described in Section III, while the initial RIS-side local geometry is obtained from compressed near-field matching:

$$\hat{\boldsymbol{\eta}}_k^{(0)} = \arg \max_{\boldsymbol{\eta} \in \mathcal{G}_k^{(0)}} \frac{|(\mathbf{h}_k^{\text{ex}}(\boldsymbol{\eta}))^H \mathbf{c}_k^{(0)}|^2}{\|\mathbf{h}_k^{\text{ex}}(\boldsymbol{\eta})\|_2^2 + \epsilon}, \quad (106)$$

where $\boldsymbol{\eta}_k = (r_k, \vartheta_k, \varphi_k)$ and

$$\mathbf{h}_k^{\text{ex}}(\boldsymbol{\eta}) = \boldsymbol{\Omega}_k [\mathbf{a}_{\text{RB},k} \odot \mathbf{a}_{\text{UR},k}^{\text{NF},\text{ex}}(\boldsymbol{\eta})]. \quad (107)$$

The common clock offset is initialized by weighted delay-geometry fusion:

$$\Delta t^{(0)} = \frac{\sum_{k=1}^K w_k^{(0)} \left[\tau_k^{(0)} - \frac{\hat{\tau}_k^{(0)} + d_{\text{RB},k}}{c} \right]}{\sum_{k=1}^K w_k^{(0)}}, \quad (108)$$

where $w_k^{(0)}$ is a confidence weight determined by the path strength or the compressed near-field correlation score in (106). A path association step is applied after initialization to align the CPD columns with the RIS panels.

C. Stage II: HP-R1P-CPD Structured Refinement

Starting from the Stage-I initialization, HP-R1P-CPD alternates between structured least-squares updates and mode-specific physical-domain rank-one projections. The EVS and RIS modes are updated by regularized LS followed by their corresponding physical projections, whereas the two delay modes are updated jointly through a common-pole ES-style structured LS subproblem in the mother delay factor $\mathbf{D}_d$. This distinction is important because the two delay factors share the same pole and appear bilinearly in the CPD model.

For the EVS and RIS modes, an unconstrained regularized LS update is first computed and then projected onto the corresponding physical manifold. For a generic mode- n factor $\mathbf{F}_n$, the regularized least-squares update is written as

$$\tilde{\mathbf{F}}_n = \arg \min_{\mathbf{F}_n} \|\mathbf{Z}_{(n)} - \mathbf{F}_n \mathbf{G}_n^T\|_F^2 + \lambda_n \|\mathbf{F}_n\|_F^2, \quad (109)$$

where $\mathbf{Z}_{(n)}$ is the mode- n unfolding of $\mathcal{Z}$, and $\mathbf{G}_n$ is formed by the Khatri-Rao product of the other factor matrices and the gain vector. In closed form,

$$\tilde{\mathbf{F}}_n = \mathbf{Z}_{(n)} \mathbf{G}_n^* (\mathbf{G}_n^T \mathbf{G}_n^* + \lambda_n \mathbf{I})^{-1}. \quad (110)$$

The regularization constants $\lambda_n > 0$ stabilize the update when the Khatri-Rao matrix is ill-conditioned or when two paths have closely spaced delays.

1) *Mode-1 EVS Projection:* For the k -th unconstrained EVS factor $\tilde{\mathbf{a}}_k$, reshape it as

$$\tilde{\mathbf{A}}_{\text{EVS},k} = \text{mat}_{6 \times M_A}(\tilde{\mathbf{a}}_k). \quad (111)$$

Since the RIS-BS direction is known from the infrastructure geometry, the BS spatial steering vector $\mathbf{v}_{\text{B},k}$ and the Maxwell matrix $\boldsymbol{\Theta}_k$ are fixed. The preliminary polarization vector is obtained by removing the known spatial response:

$$\tilde{\mathbf{p}}_k = \frac{\tilde{\mathbf{A}}_{\text{EVS},k} \mathbf{v}_{\text{B},k}^*}{\|\mathbf{v}_{\text{B},k}\|_2^2 + \epsilon}. \quad (112)$$

The Maxwell-subspace projection is

$$\bar{\mathbf{p}}_k = \mathbf{P}_{\boldsymbol{\Theta}_k} \tilde{\mathbf{p}}_k, \quad \mathbf{P}_{\boldsymbol{\Theta}_k} = \boldsymbol{\Theta}_k (\boldsymbol{\Theta}_k^H \boldsymbol{\Theta}_k + \epsilon \mathbf{I})^{-1} \boldsymbol{\Theta}_k^H. \quad (113)$$

The corresponding two-dimensional polarization coefficient is

$$\tilde{\mathbf{e}}_k = (\boldsymbol{\Theta}_k^H \boldsymbol{\Theta}_k + \epsilon \mathbf{I})^{-1} \boldsymbol{\Theta}_k^H \tilde{\mathbf{p}}_k. \quad (114)$$

To fix the scalar phase ambiguity, the phase reference is selected as

$$i_{\text{ref}} = \arg \max_{i \in \{1,2\}} |\tilde{\mathbf{e}}_k[i]|. \quad (115)$$

The phase-normalized coefficient is

$$\bar{\mathbf{e}}_k = \tilde{\mathbf{e}}_k e^{-j\angle \tilde{\mathbf{e}}_k[i_{\text{ref}}]}. \quad (116)$$

In the nominal case $|\tilde{\mathbf{e}}_k[2]|$ is sufficiently large and the second component is used as the reference to match the convention in (58). If $|\tilde{\mathbf{e}}_k[2]|$ is close to zero, the first nonzero component is used instead to avoid instability when $\gamma_k \approx \pi/2$.

The polarization parameters are then extracted according to the convention

$$\hat{\gamma}_k = \tan^{-1} \frac{|\tilde{\mathbf{e}}_k[1]|}{|\tilde{\mathbf{e}}_k[2]| + \epsilon}, \quad \hat{\eta}_k = \angle \frac{\tilde{\mathbf{e}}_k[1]}{|\tilde{\mathbf{e}}_k[2]| + \epsilon}. \quad (117)$$

The projected EVS factor is reconstructed as

$$\hat{\mathbf{a}}_k = \mathbf{v}_{\text{B},k} \otimes \boldsymbol{\Theta}_k \mathbf{e}(\hat{\gamma}_k, \hat{\eta}_k), \quad \hat{\mathbf{a}}_k \leftarrow \frac{\hat{\mathbf{a}}_k}{\|\hat{\mathbf{a}}_k\|_2}. \quad (118)$$

2) *Mode-2/3 Common-Pole ES-Style Structured LS Projection*: The unconstrained separate LS updates of $\mathbf{B}$ and $\mathbf{Q}$ do not fully exploit the fact that the two delay factors are generated by the same physical delay pole. Therefore, we do not update $\mathbf{B}$ and $\mathbf{Q}$ independently and then apply a post-processing Hankel projection. Instead, we directly update a mother delay factor $\mathbf{D}_d$, from which both delay factors are selected as

$$\mathbf{B} = \mathbf{J}_P \mathbf{D}_d, \quad \mathbf{Q} = \mathbf{J}_L \mathbf{D}_d, \quad \mathbf{D}_d \in \mathbb{C}^{R \times K}, \quad R = \max(P, L). \quad (119)$$

Because $\mathbf{B}$ and $\mathbf{Q}$ appear bilinearly in the same CPD term, the exact minimization with respect to $\mathbf{D}_d$ is not a linear LS problem. We therefore use a Gauss–Seidel structured LS approximation: at each outer iteration, the counterpart delay factor and the other tensor factors are fixed, and a linearized structured subproblem is solved for $\mathbf{D}_d$.

At the s -th outer iteration, define the mode-2 and mode-3 unfoldings as

$$\mathbf{Y}_B = \mathbf{Z}_{(2)} \in \mathbb{C}^{P \times ILT}, \quad \mathbf{Y}_Q = \mathbf{Z}_{(3)} \in \mathbb{C}^{L \times IPT}. \quad (120)$$

With $\mathbf{A}^{(s+1)}$, $\mathbf{C}^{(s)}$, $\mathbf{Q}^{(s)}$, $\mathbf{B}^{(s)}$, and $\beta^{(s)}$ fixed, the two linearized counterpart matrices are

$$\mathbf{E}_B^{(s)} = \text{Diag}(\beta^{(s)}) \left(\mathbf{C}^{(s)} \odot \mathbf{Q}^{(s)} \odot \mathbf{A}^{(s+1)} \right)^T, \quad (121)$$

and

$$\mathbf{E}_Q^{(s)} = \text{Diag}(\beta^{(s)}) \left(\mathbf{C}^{(s)} \odot \mathbf{B}^{(s)} \odot \mathbf{A}^{(s+1)} \right)^T. \quad (122)$$

The common-pole ES-style structured LS subproblem is formulated as

$$\min_{\mathbf{D}_d} h_d(\mathbf{D}_d) = \left\| \mathbf{J}_P \mathbf{D}_d \mathbf{E}_B^{(s)} - \mathbf{Y}_B \right\|_F^2 + \left\| \mathbf{J}_L \mathbf{D}_d \mathbf{E}_Q^{(s)} - \mathbf{Y}_Q \right\|_F^2 + \lambda_d \left\| \mathbf{D}_d - \mathbf{D}_d^{(s)} \right\|_F^2, \quad (123a)$$

$$\text{s.t. } \mathcal{F}(\mathbf{D}_d) \in \mathcal{T}_d. \quad (123b)$$

Here, $\mathcal{F}(\cdot)$ denotes the columnwise Hankel lifting and $\mathcal{T}_d$ denotes the frontal-rank-one tensor set. The proximal term with $\lambda_d > 0$ stabilizes the Gauss–Seidel update and prevents excessive deviation from the previous delay estimate.

Using the Wirtinger gradient up to an immaterial constant factor, the gradient of $h_d(\mathbf{D}_d)$ is

$$\nabla h_d(\mathbf{D}_d) = \mathbf{J}_P^H \left(\mathbf{J}_P \mathbf{D}_d \mathbf{E}_B^{(s)} - \mathbf{Y}_B \right) \left(\mathbf{E}_B^{(s)} \right)^H + \mathbf{J}_L^H \left(\mathbf{J}_L \mathbf{D}_d \mathbf{E}_Q^{(s)} - \mathbf{Y}_Q \right) \left(\mathbf{E}_Q^{(s)} \right)^H + \lambda_d \left(\mathbf{D}_d - \mathbf{D}_d^{(s)} \right). \quad (124)$$

Since $\mathbf{J}_P$ and $\mathbf{J}_L$ are selection matrices, a stable step size can be chosen as

$$\mu_d = \frac{\rho_d}{\left\| \mathbf{E}_B^{(s)} \left(\mathbf{E}_B^{(s)} \right)^H \right\|_2 + \left\| \mathbf{E}_Q^{(s)} \left(\mathbf{E}_Q^{(s)} \right)^H \right\|_2 + \lambda_d}, \quad 0 < \rho_d \leq 1. \quad (125)$$

The structured LS problem in (122) is solved by ES-style projected gradient descent. Starting from $\mathbf{D}_d^{(0)} = \mathbf{D}_d^{(s)}$, the

r -th inner iteration first performs an original-domain gradient step:

$$\mathbf{U}_d^{(r)} = \mathbf{D}_d^{(r)} - \mu_d \nabla h_d \left(\mathbf{D}_d^{(r)} \right). \quad (126)$$

Then the result is mapped to the Hankel domain:

$$\mathcal{U}_d^{(r)} = \mathcal{F} \left(\mathbf{U}_d^{(r)} \right) = \left[\mathcal{H}(\mathbf{u}_1^{(r)}), \dots, \mathcal{H}(\mathbf{u}_K^{(r)}) \right]. \quad (127)$$

For each frontal slice, a best rank-one approximation is applied:

$$\bar{\mathcal{U}}_d^{(r)}(:, :, k) = \Pi_{r1} \left(\mathcal{U}_d^{(r)}(:, :, k) \right) = \sigma_{1,k} \mathbf{u}_{1,k} \mathbf{v}_{1,k}^H. \quad (128)$$

The projected tensor is then mapped back to the original delay-factor domain by inverse Hankel averaging:

$$\bar{\mathbf{D}}_d^{(r+1)} = \mathcal{F}^\dagger \left(\bar{\mathcal{U}}_d^{(r)} \right), \quad (129)$$

where $\mathcal{F}^\dagger$ is anti-diagonal averaging rather than a generic matrix pseudo-inverse. Specifically, if $[\mathcal{X}]_{p,\ell,k}$ denotes the (p, ℓ) -th entry of the k -th lifted Hankel slice, then

$$[\mathcal{F}^\dagger(\mathcal{X})]_{n,k} = \frac{1}{\rho_n} \sum_{p+\ell-1=n} [\mathcal{X}]_{p,\ell,k}, \quad (130)$$

where ρ_n is the number of entries on the n -th anti-diagonal. A damped update is finally used to improve numerical stability:

$$\mathbf{D}_d^{(r+1)} = (1 - \alpha_d) \mathbf{D}_d^{(r)} + \alpha_d \bar{\mathbf{D}}_d^{(r+1)}, \quad 0 < \alpha_d \leq 1. \quad (131)$$

After R_d inner iterations, a unit-circle pole correction is applied to each column. Let $\bar{\mathbf{d}}_k$ denote the k -th column of the output mother factor. We compute

$$s_k = \sum_{r=1}^{R-1} [\bar{\mathbf{d}}_k]_{r+1} [\bar{\mathbf{d}}_k]_r^*, \quad \hat{z}_k = \frac{s_k}{|s_k| + \epsilon}. \quad (132)$$

The projected mother delay vector is reconstructed as

$$\hat{\mathbf{d}}_k = [1, \hat{z}_k, \dots, \hat{z}_k^{R-1}]^T, \quad k = 1, \dots, K. \quad (133)$$

The two delay factors are then obtained by selection:

$$\hat{\mathbf{B}} = \mathbf{J}_P \hat{\mathbf{D}}_d, \quad \hat{\mathbf{Q}} = \mathbf{J}_L \hat{\mathbf{D}}_d. \quad (134)$$

Column normalization may be applied to $\hat{\mathbf{B}}$ and $\hat{\mathbf{Q}}$, with the removed scales absorbed into the path gains.

The delay obtained from the ES-style structured LS step should not be overwritten aggressively by geometry consistency. We first compute

$$\tilde{\tau}_k = -\frac{\angle \hat{z}_k}{2\pi \Delta f}. \quad (135)$$

Using the current RIS-side range estimate $\hat{r}_k$, the common clock offset is updated as

$$\hat{\Delta t} = \frac{\sum_{k=1}^K w_k \left[\tilde{\tau}_k - \frac{\hat{r}_k + d_{\text{RB},k}}{c} \right]}{\sum_{k=1}^K w_k}. \quad (136)$$

The geometry-predicted delay is

$$\tau_k^{\text{geo}} = \frac{\hat{r}_k + d_{\text{RB},k}}{c} + \hat{\Delta t}. \quad (137)$$

A damped geometry-consistency correction is then used:

$$\hat{\tau}_k = (1 - \rho_g) \tilde{\tau}_k + \rho_g \tau_k^{\text{geo}}, \quad 0 \leq \rho_g \leq 1. \quad (138)$$

In practice, ρ_g is chosen small, e.g., $\rho_g \in [0.1, 0.3]$, or the correction is accepted only when it does not increase the tensor-domain residual. The final delay factors are reconstructed as

$$\bar{z}_k = e^{-j2\pi\Delta f\hat{\tau}_k}, \quad \hat{\mathbf{b}}_k = [1, \bar{z}_k, \dots, \bar{z}_k^{P-1}]^T, \quad \hat{\mathbf{q}}_k = [1, \bar{z}_k, \dots, \bar{z}_k^{L-1}]^T \quad (139)$$

This correction is only a tensor-stage geometry-consistency step. The statistically consistent enforcement of the single-UE geometry is performed later in the raw-domain VP-WNLS refinement.

3) *CPD Column Association for RIS-Specific Projection:* The fourth-mode projection is RIS-specific because the k -th physical RIS panel uses its own training matrix $\mathbf{\Omega}_k$ and known RIS-BS response $\mathbf{a}_{\text{RB},k}$. However, CPD has intrinsic permutation ambiguity, and the j -th algebraic column of $(\mathbf{A}, \mathbf{B}, \mathbf{Q}, \mathbf{C})$ is not automatically associated with the k -th RIS panel. Therefore, before the Mode-4 projection, we explicitly align the CPD columns with RIS panels by solving a linear assignment problem.

Let $\tilde{\mathbf{c}}_j$ be the j -th updated fourth-mode column before RIS projection. For the k -th RIS panel, define the exact-spherical compressed atom

$$\mathbf{h}_k^{\text{ex}}(\boldsymbol{\eta}) = \mathbf{\Omega}_k \left[\mathbf{a}_{\text{RB},k} \odot \mathbf{a}_{\text{UR},k}^{\text{NF,ex}}(\boldsymbol{\eta}) \right]. \quad (140)$$

The matching cost between algebraic column j and physical RIS panel k is

$$d_{j,k} = \min_{\boldsymbol{\eta} \in \mathcal{G}_k} \frac{\|\tilde{\mathbf{c}}_j - \alpha_{j,k}(\boldsymbol{\eta}) \mathbf{h}_k^{\text{ex}}(\boldsymbol{\eta})\|_2^2}{\|\tilde{\mathbf{c}}_j\|_2^2 + \epsilon}, \quad (141)$$

where

$$\alpha_{j,k}(\boldsymbol{\eta}) = \frac{(\mathbf{h}_k^{\text{ex}}(\boldsymbol{\eta}))^H \tilde{\mathbf{c}}_j}{\|\mathbf{h}_k^{\text{ex}}(\boldsymbol{\eta})\|_2^2 + \epsilon}. \quad (142)$$

The optimal column-to-panel assignment is obtained as

$$\pi^* = \arg \min_{\pi \in \mathfrak{S}_K} \sum_{k=1}^K d_{\pi(k),k}, \quad (143)$$

where $\mathfrak{S}_K$ is the set of all permutations of $\{1, \dots, K\}$. In implementation, (??) is solved by the Hungarian algorithm. After obtaining π^* , all CPD columns are reordered as

$$\mathbf{A}(:, k) \leftarrow \mathbf{A}(:, \pi^*(k)), \quad \mathbf{B}(:, k) \leftarrow \mathbf{B}(:, \pi^*(k)), \quad \mathbf{Q}(:, k) \leftarrow \mathbf{Q}(:, \pi^*(k)), \quad \mathbf{C}(:, k) \leftarrow \mathbf{C}(:, \pi^*(k)), \quad (144)$$

This assignment is performed at initialization and before each RIS-specific Mode-4 projection. It prevents the algorithm from applying the k -th RIS training matrix and RIS-BS response to an incorrectly associated CPD column.

4) *Mode-4 Physically Anchored Compressed Near-Field Projection:* The fourth-mode factor is updated by regularized least squares to obtain $\tilde{\mathbf{c}}_k \in \mathbb{C}^T$. Since $\tilde{\mathbf{c}}_k$ is a compressed training-domain factor, the near-field rank-one structure should not be imposed directly on $\mathcal{H}(\tilde{\mathbf{c}}_k)$. Instead, after the column association step in (??), we use the physically anchored compressed near-field dechirped rank-one operator $\mathcal{T}_{k,\boldsymbol{\eta}}(\mathbf{X})$ defined in Section III and solve

$$\hat{\mathbf{X}}_k(\boldsymbol{\eta}) = \arg \min_{\text{rank}(\mathbf{X})=1} \|\tilde{\mathbf{c}}_k - \mathcal{T}_{k,\boldsymbol{\eta}}(\mathbf{X})\|_2^2 + \lambda_{\text{phys}} \|\mathbf{X} - \mathbf{X}_{\boldsymbol{\eta}}^{\text{phys}}\|_F^2 \quad (145)$$

for candidate local geometries $\boldsymbol{\eta} = (r, \vartheta, \varphi)$ in a search set $\mathcal{G}_k$ centered around the current estimate. Here,

$$\mathbf{X}_{\boldsymbol{\eta}}^{\text{phys}} = \mathbf{f}_r(\boldsymbol{\eta}) \mathbf{f}_c^T(\boldsymbol{\eta}) \quad (146)$$

is the Fresnel-generated dechirped rank-one matrix for the fixed candidate $\boldsymbol{\eta}$. This anchoring term prevents the rank-one lifted variable from becoming an arbitrary nonphysical rank-one matrix. The lifted variable is used only as a local denoising surrogate for candidate generation, while the final RIS response is reconstructed by the exact spherical model.

For a fixed $\boldsymbol{\eta}$, write $\mathcal{T}_{k,\boldsymbol{\eta}}(\mathbf{X}) = \mathbf{T}_{k,\boldsymbol{\eta}} \text{vec}(\mathbf{X})$, $\mathbf{x} = \text{vec}(\mathbf{X})$, and $\mathbf{x}_{\boldsymbol{\eta}}^{\text{phys}} = \text{vec}(\mathbf{X}_{\boldsymbol{\eta}}^{\text{phys}})$. The rank-one fitting in (139) is initialized by $\mathbf{X}^{(0)} = \mathbf{X}_{\boldsymbol{\eta}}^{\text{phys}}$ and solved by a small number of projected-gradient iterations:

$$\mathbf{x}^{(\ell+\frac{1}{2})} = \mathbf{x}^{(\ell)} + \mu_c \left[\mathbf{T}_{k,\boldsymbol{\eta}}^H \left(\tilde{\mathbf{c}}_k - \mathbf{T}_{k,\boldsymbol{\eta}} \mathbf{x}^{(\ell)} \right) - \lambda_{\text{phys}} \left(\mathbf{x}^{(\ell)} - \mathbf{x}_{\boldsymbol{\eta}}^{\text{phys}} \right) \right], \quad (147)$$

$$\mathbf{X}^{(\ell+1)} = \Pi_{\text{r1}} \left(\text{mat} \left(\mathbf{x}^{(\ell+\frac{1}{2})} \right) \right), \quad 0 < \mu_c < \frac{1}{\|\mathbf{T}_{k,\boldsymbol{\eta}}\|_2^2 + \lambda_{\text{phys}}}. \quad (148)$$

The number of inner iterations is intentionally kept small so that the lifted variable performs local structured denoising around the Fresnel-generated physical anchor rather than fully relaxing the near-field manifold.

The regularized compressed-domain residual for candidate $\boldsymbol{\eta}$ is

$$\rho_k(\boldsymbol{\eta}) = \left\| \tilde{\mathbf{c}}_k - \mathcal{T}_{k,\boldsymbol{\eta}}(\hat{\mathbf{X}}_k(\boldsymbol{\eta})) \right\|_2^2 + \lambda_{\text{phys}} \left\| \hat{\mathbf{X}}_k(\boldsymbol{\eta}) - \mathbf{X}_{\boldsymbol{\eta}}^{\text{phys}} \right\|_F^2. \quad (149)$$

The Fresnel-domain candidate is selected by

$$\hat{\boldsymbol{\eta}}_k^{\text{F}} = \arg \min_{\boldsymbol{\eta} \in \mathcal{G}_k} \rho_k(\boldsymbol{\eta}). \quad (150)$$

The exact spherical model is then used only as a local physical refinement initialized by $\hat{\boldsymbol{\eta}}_k^{\text{F}}$. Define

$$\mathbf{h}_k^{\text{ex}}(\boldsymbol{\eta}) = \mathbf{\Omega}_k \left[\mathbf{a}_{\text{RB},k} \odot \mathbf{a}_{\text{UR},k}^{\text{NF,ex}}(\boldsymbol{\eta}) \right]. \quad (151)$$

For a fixed $\boldsymbol{\eta}$, the optimal complex scale is

$$\alpha_k(\boldsymbol{\eta}) = \frac{(\mathbf{h}_k^{\text{ex}}(\boldsymbol{\eta}))^H \tilde{\mathbf{c}}_k}{\|\mathbf{h}_k^{\text{ex}}(\boldsymbol{\eta})\|_2^2 + \epsilon}. \quad (152)$$

The local exact-spherical objective is

$$\Phi_k(\boldsymbol{\eta}) = \left\| \tilde{\mathbf{c}}_k - \alpha_k(\boldsymbol{\eta}) \mathbf{h}_k^{\text{ex}}(\boldsymbol{\eta}) \right\|_2^2. \quad (153)$$

Equivalently, it can be evaluated as

$$\Phi_k(\boldsymbol{\eta}) = \|\tilde{\mathbf{c}}_k\|_2^2 - \frac{\left| (\mathbf{h}_k^{\text{ex}}(\boldsymbol{\eta}))^H \tilde{\mathbf{c}}_k \right|^2}{\|\mathbf{h}_k^{\text{ex}}(\boldsymbol{\eta})\|_2^2 + \epsilon}. \quad (154)$$

Starting from $\hat{\boldsymbol{\eta}}_k^{\text{F}}$, the exact-spherical local refinement is

$$\hat{\boldsymbol{\eta}}_k^{\text{ex}} = \text{LocalRefine} \left(\hat{\boldsymbol{\eta}}_k^{\text{F}}, \Phi_k(\boldsymbol{\eta}) \right). \quad (155)$$

The projected fourth-mode factor is reconstructed by

$$\hat{\mathbf{c}}_k = \alpha_k(\hat{\boldsymbol{\eta}}_k^{\text{ex}}) \mathbf{h}_k^{\text{ex}}(\hat{\boldsymbol{\eta}}_k^{\text{ex}}), \quad \hat{\mathbf{c}}_k \leftarrow \frac{\hat{\mathbf{c}}_k}{\|\hat{\mathbf{c}}_k\|_2}. \quad (156)$$

The scale removed by column normalization is absorbed into the complex path gain β_k . The local range and angles stored for geometry correction are obtained from $\hat{\boldsymbol{\eta}}_k^{\text{ex}}$.

5) *Gain Update and Stopping Criterion:* After updating all structured factors, define

$$\Psi_Z = [\hat{\mathbf{c}}_1 \otimes \hat{\mathbf{q}}_1 \otimes \hat{\mathbf{b}}_1 \otimes \hat{\mathbf{a}}_1, \dots, \hat{\mathbf{c}}_K \otimes \hat{\mathbf{q}}_K \otimes \hat{\mathbf{b}}_K \otimes \hat{\mathbf{a}}_K]. \quad (157)$$

Let $\mathbf{z} = \text{vec}(\mathcal{Z})$. The path gains are estimated by regularized least squares:

$$\hat{\beta} = (\Psi_Z^H \Psi_Z + \lambda_\beta \mathbf{I}_K)^{-1} \Psi_Z^H \mathbf{z}. \quad (158)$$

The normalized residual at iteration s is

$$\varepsilon^{(s)} = \frac{\|\mathbf{z} - \Psi_Z \hat{\beta}\|_2^2}{\|\mathbf{z}\|_2^2}. \quad (159)$$

The HP-R1P-CPD iteration is stopped when

$$\frac{|\varepsilon^{(s)} - \varepsilon^{(s-1)}|}{\max(\varepsilon^{(s-1)}, \varepsilon_0)} < \varepsilon_{\text{tol}}, \quad (160)$$

or when the maximum number of iterations is reached.

D. Stage III: Raw-Domain Global VP-WNLS Refinement

The structured tensor refinement estimates path-local quantities, including $\{\hat{r}_k, \hat{\vartheta}_k, \hat{\varphi}_k, \hat{\tau}_k, \hat{\gamma}_k, \hat{\eta}_k\}_{k=1}^K$. However, in a distributed RIS system illuminated by a single UE, the local ranges and incident angles must be induced by the same global UE position. Therefore, the final stage refines the estimates under the global parameterization in (101).

From the Stage-II local estimate $\hat{\eta}_k = (\hat{r}_k, \hat{\vartheta}_k, \hat{\varphi}_k)$, one UE position candidate is generated from each RIS panel:

$$\hat{\mathbf{p}}_{u,k} = \mathbf{s}_k + \mathbf{R}_k^{\text{R} \rightarrow \text{G}} [\hat{r}_k \mathbf{u}(\hat{\vartheta}_k, \hat{\varphi}_k)]. \quad (161)$$

The global UE position is initialized by confidence-weighted fusion:

$$\hat{\mathbf{p}}_u^{(0)} = \frac{\sum_{k=1}^K w_k \hat{\mathbf{p}}_{u,k}}{\sum_{k=1}^K w_k}. \quad (162)$$

The clock offset is initialized as

$$\hat{\Delta t}^{(0)} = \frac{\sum_{k=1}^K w_k \left[\hat{\tau}_k - \frac{\|\mathbf{R}_k^{\text{G} \rightarrow \text{R}}(\hat{\mathbf{p}}_u^{(0)} - \mathbf{s}_k)\|_2 + d_{\text{RB},k}}{c} \right]}{\sum_{k=1}^K w_k}. \quad (163)$$

For a given global parameter vector χ , the local UE-RIS vector, range, and delay of the k -th path are deterministically reconstructed as

$$\mathbf{q}_k(\chi) = \mathbf{R}_k^{\text{G} \rightarrow \text{R}}(\mathbf{p}_u - \mathbf{s}_k), \quad r_k(\chi) = \|\mathbf{q}_k(\chi)\|_2, \quad (164)$$

$$\tau_k(\chi) = \frac{r_k(\chi) + d_{\text{RB},k}}{c} + \Delta t. \quad (165)$$

The raw-domain delay vector is

$$\mathbf{d}_k(\tau_k) = [1, e^{-j2\pi\Delta f\tau_k}, \dots, e^{-j2\pi(N-1)\Delta f\tau_k}]^T. \quad (166)$$

The compressed RIS factor reconstructed from the global UE position is

$$\mathbf{c}_k(\mathbf{p}_u) = \mathbf{\Omega}_k [\mathbf{a}_{\text{RB},k} \odot \mathbf{a}_{\text{UR},k}^{\text{NF}}(\mathbf{q}_k(\chi))]. \quad (167)$$

The raw-domain atom of the k -th path is

$$\psi_{Y,k}(\chi) = \mathbf{c}_k(\mathbf{p}_u) \otimes \mathbf{d}_k(\tau_k) \otimes \mathbf{a}_{\text{EV},k}(\gamma_k, \eta_k). \quad (168)$$

Stacking all atoms gives

$$\Psi_Y(\chi) = [\psi_{Y,1}(\chi), \dots, \psi_{Y,K}(\chi)]. \quad (169)$$

Let $\mathbf{y} = \text{vec}(\mathcal{Y})$ denote the raw OFDM observation vector. For a fixed χ , the complex path gains are eliminated by variable projection:

$$\hat{\beta}(\chi) = (\Psi_Y^H(\chi) \mathbf{W} \Psi_Y(\chi) + \lambda_\beta \mathbf{I}_K)^{-1} \Psi_Y^H(\chi) \mathbf{W} \mathbf{y}, \quad (170)$$

where $\mathbf{W}$ is a weighting matrix. For spatially white raw-domain noise, $\mathbf{W} = \mathbf{I}$. The final global VP-WNLS problem is

$$\hat{\chi} = \arg \min_{\chi} \left\| \mathbf{W}^{1/2} [\mathbf{y} - \Psi_Y(\chi) \hat{\beta}(\chi)] \right\|_2^2. \quad (171)$$

In implementation, the optimization is performed over normalized variables to improve numerical conditioning:

$$\mathbf{z}_\chi = \left[\frac{(\mathbf{p}_u - \hat{\mathbf{p}}_u^{(0)})^T}{s_p}, \frac{\Delta t - \hat{\Delta t}^{(0)}}{s_t}, \frac{(\gamma - \hat{\gamma}^{(0)})^T}{s_a}, \frac{(\eta - \hat{\eta}^{(0)})^T}{s_a} \right]^T, \quad (172)$$

where typical scaling constants are $s_p = 1$ m, $s_t = 1$ ns, and $s_a = 0.1$ rad. The final output consists of $\hat{\mathbf{p}}_u, \hat{\Delta t}, \{\hat{\gamma}_k, \hat{\eta}_k\}_{k=1}^K, \hat{\beta}$, and the derived local quantities $\{\hat{r}_k, \hat{\vartheta}_k, \hat{\varphi}_k, \hat{\tau}_k\}_{k=1}^K$.

E. Overall Algorithm

The complete procedure is summarized in Algorithm 1. The algorithm uses HOSVD-ESPRIT to obtain a reliable initialization, HP-R1P-CPD to refine the heterogeneous tensor factors, and raw-domain VP-WNLS to enforce global geometry and remove the bias caused by treating Hankelized samples as independent observations.

F. Computational Complexity and Implementation Remarks

Let S_{CPD} denote the number of HP-R1P-CPD iterations and let S_{VP} denote the number of raw-domain VP-WNLS iterations. The dominant cost of Stage I is the truncated HOSVD and the joint shift-invariance operation. For $\mathcal{Z} \in \mathbb{C}^{I \times P \times L \times T}$, a typical truncated-SVD implementation has leading complexity

$$\mathcal{O}(KIPLT + K^2(I + P + L + T)). \quad (173)$$

Each HP-R1P-CPD iteration contains regularized Khatri-Rao least-squares updates with leading cost

$$\mathcal{O}(KIPLT + K^2(I + P + L + T)), \quad (174)$$

excluding the RIS near-field projection and the column association step.

The Mode-4 projection is the most expensive structured projection because it evaluates compressed near-field candidates and performs physically anchored rank-one projected-gradient fitting. If N_{cand} is the number of local geometry candidates and the dechirped lifted matrix has size $P_x P_y \times L_x L_y$, the approximate per-iteration cost of the RIS projection is

$$\mathcal{O}(KN_{\text{cand}}[TM_R + P_x P_y L_x L_y + C_{\text{svd}}]), \quad (175)$$

where $C_{\text{svd}} \approx P_x P_y L_x L_y$ when a rank-one partial SVD is used. The column-to-RIS association step requires constructing a $K \times K$ cost matrix and solving a linear assignment problem. Its complexity is approximately

$$\mathcal{O}(K^2 N_{\text{assn}} T M_R + K^3), \quad (176)$$

where N_{assn} is the number of local candidates used in the assignment score. Since K is typically small in sparse RIS-assisted channels, this step adds limited overhead but removes the otherwise dangerous permutation ambiguity before RIS-specific projection.

The raw-domain VP-WNLS stage constructs $\Psi_Y(\chi) \in \mathbb{C}^{INT \times K}$ and solves a $K \times K$ regularized linear system at each iteration. The dominant complexity is

$$\mathcal{O}(S_{\text{VP}} [KINT + K^2 INT + K^3]). \quad (177)$$

Since the final refinement operates on the raw OFDM samples instead of duplicated Hankelized samples, it avoids overcounting repeated observations and provides a more appropriate likelihood-based refinement. The HP-RIP-CPD stage should be interpreted as a safeguarded alternating structured projection method initialized by HOSVD-ESPRIT. It is designed for stable local convergence and physical consistency, but no global optimality claim is made.

V. IDENTIFIABILITY ANALYSIS

This section discusses the identifiability of the proposed mixed-field distributed RIS-EVS-OFDM tensor model. The analysis is divided into two layers. First, we establish the essential uniqueness of the fourth-order CPD factors. Second, we discuss the local identifiability of the underlying physical parameters from the identifiable tensor factors and from the raw-domain global model.

A. Essential Uniqueness of the Fourth-Order CPD

Recall that the Hankelized tensor observation is modeled as

$$\mathcal{Z} = \sum_{k=1}^K \beta_k \mathbf{a}_{\text{EVS},k} \circ \mathbf{b}_k \circ \mathbf{q}_k \circ \mathbf{c}_k + \mathcal{W}_H, \quad (178)$$

or equivalently,

$$\mathcal{Z} = [[\beta; \mathbf{A}, \mathbf{B}, \mathbf{Q}, \mathbf{C}]] + \mathcal{W}_H, \quad (179)$$

where

$$\mathbf{A} = [\mathbf{a}_{\text{EVS},1}, \dots, \mathbf{a}_{\text{EVS},K}] \in \mathbb{C}^{I \times K}, \quad (180)$$

$$\mathbf{B} = [\mathbf{b}_1, \dots, \mathbf{b}_K] \in \mathbb{C}^{P \times K}, \quad (181)$$

$$\mathbf{Q} = [\mathbf{q}_1, \dots, \mathbf{q}_K] \in \mathbb{C}^{L \times K}, \quad (182)$$

$$\mathbf{C} = [\mathbf{c}_1, \dots, \mathbf{c}_K] \in \mathbb{C}^{T \times K}. \quad (183)$$

Here, $I = 6M_A$, $\mathbf{b}_k$ and $\mathbf{q}_k$ are two Vandermonde vectors sharing the same delay pole, and $\mathbf{c}_k = \Omega_k \mathbf{g}_k$ is the compressed RIS training-domain factor.

Let k_A , k_B , k_Q , and k_C denote the Kruskal ranks of $\mathbf{A}$, $\mathbf{B}$, $\mathbf{Q}$, and $\mathbf{C}$, respectively. The following proposition gives a sufficient condition for essential uniqueness.

Proposition 1 (Essential CPD uniqueness): Assume that the factor matrices in (170) satisfy

$$k_A + k_B + k_Q + k_C \geq 2K + 3. \quad (184)$$

Then the fourth-order CPD in (170) is essentially unique, i.e., the factor matrices are identifiable up to common column permutation and mode-dependent scaling ambiguities.

Proof: The result follows from the Kruskal-type uniqueness condition for N -way CPD. For an N -th order rank- K tensor, a sufficient condition for essential uniqueness is

$$\sum_{n=1}^N k_n \geq 2K + N - 1. \quad (185)$$

Setting $N = 4$ and identifying the four factor matrices as $(\mathbf{A}, \mathbf{B}, \mathbf{Q}, \mathbf{C})$ gives (176). The remaining ambiguities are the standard CPD permutation and scaling ambiguities. ■

Proposition 1 is a sufficient rather than necessary condition. It is nevertheless useful because it explicitly reveals how the EVS dimension, frequency smoothing dimensions, and RIS training dimension jointly support tensor identifiability.

B. Kruskal Ranks of the Structured Factors

The condition in (176) can be connected to the physical structures of the proposed model.

1) *Delay Factors:* For the k -th path, the two frequency-smoothed delay factors are

$$\mathbf{b}_k = [1, z_k, \dots, z_k^{P-1}]^T, \quad \mathbf{q}_k = [1, z_k, \dots, z_k^{L-1}]^T, \quad (186)$$

where

$$z_k = e^{-j2\pi\Delta f\tau_k}. \quad (187)$$

If the delay poles $\{z_k\}_{k=1}^K$ are distinct, then the Vandermonde matrices $\mathbf{B}$ and $\mathbf{Q}$ satisfy

$$k_B = \min(P, K), \quad k_Q = \min(L, K). \quad (188)$$

Thus, the frequency-smoothing dimensions P and L directly improve the algebraic identifiability of the delay subspace.

2) *EVS Spatial-Polarization Factor:* The BS-side factor is

$$\mathbf{a}_{\text{EVS},k} = \mathbf{v}_{\text{B},k} \otimes \Theta_k \mathbf{e}(\gamma_k, \eta_k), \quad (189)$$

where the RIS-BS direction is known from the infrastructure geometry and

$$\mathbf{e}(\gamma_k, \eta_k) = [\sin \gamma_k e^{j\eta_k}, \cos \gamma_k]^T. \quad (190)$$

If the RIS-BS directions are not spatially aliased and the polarization vectors are not mutually degenerate, then the EVS spatial-polarization factor generically satisfies

$$k_A = \min(I, K), \quad I = 6M_A. \quad (191)$$

Degenerate cases include identical BS-side spatial steering vectors combined with collinear EVS polarization responses, or polarization singularities such as $\sin \gamma_k = 0$, where the polarization phase η_k becomes unobservable.

3) *Compressed RIS Near-Field Factor*: The fourth-mode factor is

$$\mathbf{c}_k = \mathbf{\Omega}_k \mathbf{g}_k, \quad \mathbf{g}_k = \mathbf{a}_{\text{RB},k} \odot \mathbf{a}_{\text{UR},k}^{\text{NF}}. \quad (192)$$

Since $\mathbf{c}_k$ is a compressed training-domain vector, its linear independence depends jointly on the RIS training matrix and the element-domain near-field responses. If the RIS training patterns are sufficiently diverse and the compressed near-field fingerprints

$$\{\mathbf{\Omega}_k \mathbf{g}_k\}_{k=1}^K \quad (193)$$

are in general position, then

$$k_{\text{C}} = \min(T, K) \quad (194)$$

holds generically. This condition fails when the training patterns are rank deficient, when two paths generate identical compressed RIS fingerprints, or when the RIS aperture is too small for the near-field curvature to be distinguishable after compression.

Combining (180), (183), and (186), a practical generic sufficient condition for essential CPD uniqueness is

$$\min(I, K) + \min(P, K) + \min(L, K) + \min(T, K) \geq 2K + 3. \quad (195)$$

For the common case $I \geq K$, $P \geq K$, $L \geq K$, and $T \geq K$, condition (187) is automatically satisfied for $K \geq 2$. The rank-one case $K = 1$ is trivially identifiable after fixing the scaling convention.

C. Removal of Scaling and Permutation Ambiguities

Essential CPD uniqueness determines the factor matrices up to column permutation and scaling. Specifically, for each path k , the factor tuple can be transformed as

$$(\mathbf{a}_k, \mathbf{b}_k, \mathbf{q}_k, \mathbf{c}_k, \beta_k) \rightarrow (\lambda_{a,k} \mathbf{a}_k, \lambda_{b,k} \mathbf{b}_k, \lambda_{q,k} \mathbf{q}_k, \lambda_{c,k} \mathbf{c}_k, \beta_k / (\lambda_{a,k} \lambda_{b,k} \lambda_{q,k} \lambda_{c,k})), \quad (196)$$

without changing the reconstructed tensor.

These ambiguities do not affect physical parameter extraction. The delay factors are normalized by enforcing

$$[\mathbf{b}_k]_1 = [\mathbf{q}_k]_1 = 1, \quad (197)$$

which fixes their scale and allows the delay pole z_k to be extracted. The EVS factor is normalized before polarization recovery, and the remaining scalar is absorbed into β_k . The RIS training-domain factor is also normalized before near-field geometry extraction, since the local geometry is determined by the direction of $\mathbf{c}_k$ in the compressed training space rather than by its arbitrary CPD scale.

The column permutation ambiguity is resolved through path-to-RIS association. Since each dominant path is associated with a known distributed RIS panel, the recovered column can be matched to the RIS index by the compressed near-field fitting score

$$\rho_{k,\ell} = \max_{\eta, \alpha} \frac{|\tilde{\mathbf{c}}_{\ell}^H \mathbf{\Omega}_k \mathbf{g}_k(\eta)|^2}{\|\tilde{\mathbf{c}}_{\ell}\|_2^2 \|\mathbf{\Omega}_k \mathbf{g}_k(\eta)\|_2^2}, \quad (198)$$

or by an equivalent assignment rule. After association, all factor matrices and physical parameters are permuted consistently.

D. Identifiability of Path-Local Physical Parameters

After essential uniqueness of the CPD factors, the next question is whether the physical parameters can be recovered from the identifiable factors.

1) *Delay Identifiability*: Since $\mathbf{b}_k$ and $\mathbf{q}_k$ share the same pole z_k , the delay can be recovered from either factor or from a joint projection:

$$\hat{z}_k = \arg \min_{|z|=1, \alpha_b, \alpha_q} \|\hat{\mathbf{b}}_k - \alpha_b \mathbf{b}(z)\|_2^2 + \|\hat{\mathbf{q}}_k - \alpha_q \mathbf{q}(z)\|_2^2. \quad (199)$$

The corresponding delay is

$$\hat{\tau}_k = -\frac{\angle \hat{z}_k}{2\pi \Delta f}. \quad (200)$$

This recovery is unique within the prescribed unambiguous delay interval. Equivalently, if the physical delay support has length smaller than $1/\Delta f$, the mapping $\tau_k \mapsto z_k$ is one-to-one over that interval.

2) *Polarization Identifiability*: Given the identifiable EVS factor and the known RIS-BS direction, the six-dimensional polarization vector satisfies

$$\mathbf{p}_k = \mathbf{\Theta}_k \mathbf{e}(\gamma_k, \eta_k). \quad (201)$$

If $\mathbf{\Theta}_k$ has full column rank, then

$$\mathbf{e}(\gamma_k, \eta_k) = (\mathbf{\Theta}_k^H \mathbf{\Theta}_k)^{-1} \mathbf{\Theta}_k^H \mathbf{p}_k \quad (202)$$

is identifiable up to the same scalar ambiguity already absorbed by β_k . Under the non-degenerate condition

$$0 < \gamma_k < \frac{\pi}{2}, \quad (203)$$

the polarization parameters are locally identifiable from the ratio of the two entries:

$$\gamma_k = \tan^{-1} \frac{|e_{k,1}|}{|e_{k,2}|}, \quad \eta_k = \angle \frac{e_{k,1}}{e_{k,2}}. \quad (204)$$

3) *RIS Near-Field Geometry Identifiability*: For the RIS near-field factor, define the compressed near-field mapping

$$\mathbf{h}_k(\eta_k) = \mathbf{\Omega}_k [\mathbf{a}_{\text{RB},k} \odot \mathbf{a}_{\text{UR},k}^{\text{NF}}(\eta_k)], \quad (205)$$

where

$$\eta_k = [r_k, \vartheta_k^{\text{R}}, \varphi_k^{\text{R}}]^T. \quad (206)$$

Because the scale of $\mathbf{c}_k$ is absorbed by the complex gain, the relevant geometry is identified from the normalized compressed fingerprint. Let

$$\mathbf{J}_{\eta,k} = \begin{bmatrix} \frac{\partial \mathbf{h}_k}{\partial r_k}, \frac{\partial \mathbf{h}_k}{\partial \vartheta_k^{\text{R}}}, \frac{\partial \mathbf{h}_k}{\partial \varphi_k^{\text{R}}} \end{bmatrix} \quad (207)$$

be the complex Jacobian of $\mathbf{h}_k$ with respect to the local RIS-side geometry, and define the orthogonal projector

$$\mathbf{P}_{\mathbf{h}_k}^{\perp} = \mathbf{I}_T - \mathbf{h}_k (\mathbf{h}_k^H \mathbf{h}_k)^{-1} \mathbf{h}_k^H. \quad (208)$$

Then the local RIS-side geometry is locally identifiable, up to the scalar ambiguity, if

$$\text{rank}_{\mathbb{R}} \left(\begin{bmatrix} \Re \{ \mathbf{P}_{\mathbf{h}_k}^{\perp} \mathbf{J}_{\eta,k} \} \\ \Im \{ \mathbf{P}_{\mathbf{h}_k}^{\perp} \mathbf{J}_{\eta,k} \} \end{bmatrix} \right) = 3. \quad (209)$$

Condition (201) states that the compressed training observations must contain three independent real directions for resolving range, elevation, and azimuth after removing the irrelevant complex scaling direction. It requires sufficiently diverse RIS training patterns and a non-degenerate near-field geometry. It may fail in the far-field limit, for insufficient RIS aperture, for rank-deficient training patterns, or when different local geometries produce indistinguishable compressed fingerprints.

E. Global Geometry Identifiability

The path-local parameters are intermediate quantities. The final physical parameter vector is

$$\chi = [\mathbf{p}_u^T, \Delta t, \gamma_1, \dots, \gamma_K, \eta_1, \dots, \eta_K]^T. \quad (210)$$

Given a local geometry estimate from RIS k , the UE position satisfies

$$\mathbf{p}_u = \mathbf{s}_k + \mathbf{R}_k^{\text{R} \rightarrow \text{G}} [r_k \mathbf{u}(\vartheta_k^{\text{R}}, \varphi_k^{\text{R}})]. \quad (211)$$

Thus, if at least one RIS provides a non-degenerate identifiable triplet $(r_k, \vartheta_k^{\text{R}}, \varphi_k^{\text{R}})$ and its local frame is known, $\mathbf{p}_u$ is locally identifiable from (203). Once $\mathbf{p}_u$ and the path delay are identifiable, the common clock offset is obtained from

$$\Delta t = \tau_k - \frac{r_k + d_{\text{RB},k}}{c}. \quad (212)$$

Multiple distributed RIS panels provide redundant constraints and improve robustness against noise and local degeneracy.

A more compact and statistically relevant condition can be stated directly in the raw observation domain. Let

$$\mathbf{y} = \Psi_Y(\chi) \beta + \mathbf{w} \quad (213)$$

be the raw-domain model, and let

$$\mathbf{D}_\chi = \frac{\partial \mu(\chi, \beta)}{\partial \chi} \quad (214)$$

with

$$\mu(\chi, \beta) = \Psi_Y(\chi) \beta. \quad (215)$$

Since the complex gains are unknown nuisance parameters, perturbations lying in the column space of Ψ_Y can be absorbed by gain changes. Therefore, the effective local identifiability condition is

$$\text{rank}_{\mathbb{R}} \left(\begin{bmatrix} \Re\{\mathbf{P}_\Psi^\perp \mathbf{D}_\chi\} \\ \Im\{\mathbf{P}_\Psi^\perp \mathbf{D}_\chi\} \end{bmatrix} \right) = 4 + 2K, \quad (216)$$

where

$$\mathbf{P}_\Psi^\perp = \mathbf{I} - \Psi_Y(\Psi_Y^H \Psi_Y)^\dagger \Psi_Y^H. \quad (217)$$

Condition (208) is also the nonsingularity condition of the equivalent Fisher information matrix after eliminating the unknown complex gains.

Proposition 2 (Local identifiability of the global physical parameters): Assume that the CPD factors in (170) are essentially identifiable, the delay poles are unambiguous, the EVS polarization parameters are non-degenerate, and the compressed RIS near-field mappings satisfy (201). If the raw-domain projected Jacobian satisfies (208), then the global parameter vector

$$\chi = [\mathbf{p}_u^T, \Delta t, \gamma_1, \dots, \gamma_K, \eta_1, \dots, \eta_K]^T$$

is locally identifiable after eliminating the unknown complex path gains.

Proof: By Proposition 1, the tensor factors are identifiable up to permutation and scaling. The scaling ambiguities are removed by the normalization conventions, and the permutation ambiguity is resolved through path-to-RIS association. The common-delay Vandermonde factors identify the delay poles within the unambiguous delay interval. The EVS factor identifies the polarization parameters under the full-column-rank Maxwell mapping and nonzero polarization components. The compressed RIS near-field factor identifies the local RIS-side geometry when (201) holds. Finally, the global parameters are locally identifiable if no nonzero perturbation of χ produces a mean variation that can be absorbed by the unknown complex gains. This is exactly the projected-Jacobian full-rank condition in (208). ■

F. Discussion

The above conditions should be interpreted as sufficient and local identifiability conditions rather than as necessary conditions. They clarify the roles of the main system dimensions and physical assumptions:

- The frequency-smoothing dimensions P and L improve identifiability through the two common-delay Vandermonde factors.
- The EVS dimension $I = 6M_A$ improves the Kruskal rank of the BS-side spatial-polarization factor and enables polarization recovery.
- The RIS training length T controls the rank of the compressed near-field factor $\mathbf{C}$. A small T may still work with strong physical priors, but algebraic CPD identifiability requires sufficient diversity in $\{\mathbf{c}_k\}$.
- The RIS aperture and the UE-RIS distance determine whether the near-field curvature provides independent range information. In the far-field limit, the range derivative becomes weak and the local near-field geometry may become ill-conditioned.
- The raw-domain projected-Jacobian condition (208) links identifiability to the same parameterization used by the final VP-WNLS refinement and the CRB analysis.

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Algorithm 1 HP-R1P-CPD With Raw-Domain VP-WNLS

Require: Raw OFDM observation $\mathcal{Y}$, Hankelized tensor $\mathcal{Z}$, RIS training matrices $\{\Omega_k\}_{k=1}^K$, known RIS-BS responses $\{\mathbf{a}_{\text{RB},k}\}_{k=1}^K$, system geometry, rank K , and stopping tolerance ε_{tol} .

Ensure: Global UE position $\hat{\mathbf{p}}_u$, clock offset $\hat{\Delta t}$, polarization parameters $\{\hat{\eta}_k, \hat{\eta}_k\}_{k=1}^K$, path gains $\hat{\beta}$, and derived local parameters.

- 1: Compute a truncated HOSVD of $\mathcal{Z}$.
 - 2: Estimate the initial delay poles $\{z_k^{(0)}\}_{k=1}^K$ from the joint shift-invariance of modes 2 and 3.
 - 3: Reconstruct $\mathbf{B}^{(0)}$ and $\mathbf{Q}^{(0)}$ from $\{z_k^{(0)}\}_{k=1}^K$.
 - 4: Extract $\mathbf{A}^{(0)}$ and $\mathbf{C}^{(0)}$ by regularized least squares and rank-one SVD separation.
 - 5: Project $\mathbf{A}^{(0)}$ onto the EVS Maxwell-Kronecker manifold.
 - 6: Obtain coarse local RIS near-field estimates by compressed near-field matching.
 - 7: Solve the initial column-to-RIS association problem in (??) and reorder the CPD columns.
 - 8: Initialize $\Delta t^{(0)}$ by delay-geometry fusion.
 - 9: **repeat**
 - 10: Update $\tilde{\mathbf{A}}$ by regularized LS and project each column onto the EVS Maxwell-Kronecker manifold.
 - 11: Construct the linearized counterpart matrices $\mathbf{E}_B^{(s)}$ and $\mathbf{E}_Q^{(s)}$ using the current factors.
 - 12: Update the mother delay factor $\mathbf{D}_d$ by solving the common-pole ES-style structured LS subproblem in (122) using projected gradient iterations.
 - 13: Apply Hankel-domain frontal-rank-one projection, inverse Hankel averaging, and unit-circle pole correction to obtain $\hat{\mathbf{D}}_d$.
 - 14: Reconstruct $\hat{\mathbf{B}} = \mathbf{J}_P \hat{\mathbf{D}}_d$ and $\hat{\mathbf{Q}} = \mathbf{J}_L \hat{\mathbf{D}}_d$.
 - 15: Apply the damped geometry-consistency correction in (137) if it does not increase the tensor-domain residual.
 - 16: Update $\tilde{\mathbf{C}}$ by regularized LS.
 - 17: Solve the column-to-RIS assignment problem in (??) and reorder $\mathbf{A}$, $\mathbf{B}$, $\mathbf{Q}$, $\mathbf{C}$, and β .
 - 18: **for** $k = 1, \dots, K$ **do**
 - 19: Form a local candidate set $\mathcal{G}_k$ around the current RIS-side geometry estimate.
 - 20: For each $\eta \in \mathcal{G}_k$, initialize $\mathbf{X}^{(0)} = \mathbf{X}_\eta^{\text{phys}}$ and solve the physically anchored compressed near-field fitting problem in (139) using a few projected-gradient iterations.
 - 21: Select the best Fresnel candidate $\hat{\eta}_k^{\text{F}}$ according to (143).
 - 22: Perform exact-spherical local physical refinement initialized from $\hat{\eta}_k^{\text{F}}$.
 - 23: Reconstruct and normalize the projected fourth-mode factor $\hat{\mathbf{c}}_k$.
 - 24: **end for**
 - 25: Recover $\hat{\beta}$ by (151).
 - 26: Compute the normalized residual $\varepsilon^{(s)}$ by (152).
 - 27: **until** the stopping condition in (153) is satisfied or the maximum number of iterations is reached
 - 28: Fuse the Stage-II local estimates to initialize $\hat{\mathbf{p}}_u^{(0)}$ and $\hat{\Delta t}^{(0)}$.
 - 29: Solve the raw-domain VP-WNLS problem in (164).
 - 30: Recover $\hat{\beta}$ by (163).
 - 31: Derive $\{\hat{r}_k, \hat{\vartheta}_k, \hat{\varphi}_k, \hat{\tau}_k\}_{k=1}^K$ from $\hat{\chi}$.
-